

Market Commentary | December 14, 2017

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Ginnie Mae Project Loan Modeling

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Supporting Affordable Housing



Source: PEXELS.

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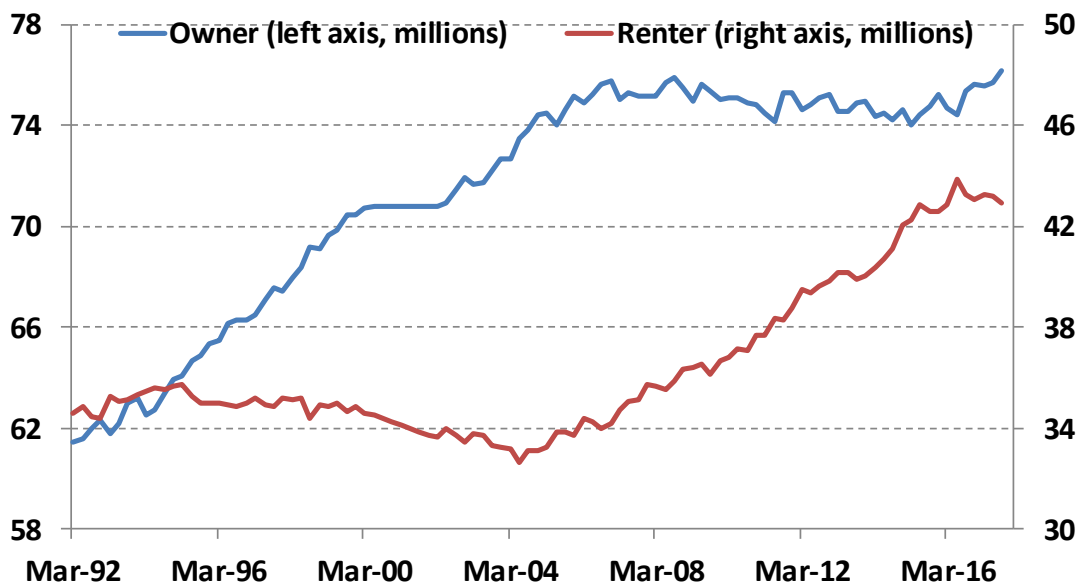
Introduction

Multifamily Housing's Post-Recession Growth

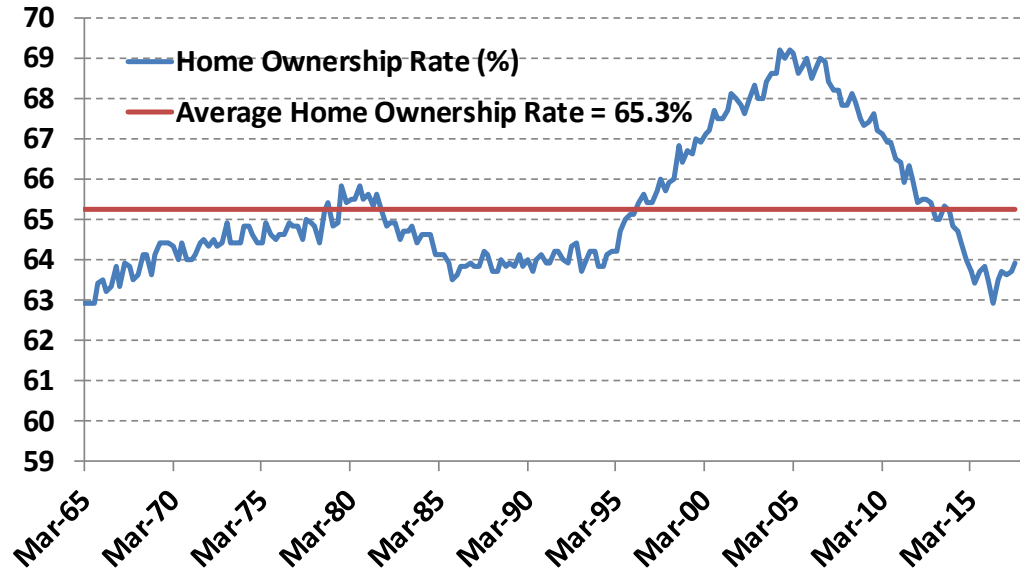
Background: Economic Stress and City-Living Drive Renting

The realities of a weak income growth economy in combination with a preference for living in cities has continued to drive households to rent rather than own, resulting in a homeownership rate that rapidly fell post-recession and is only now inching upwards (see Figure 1).

Figure 1. Renting the New Owning (Top) Pushes Homeownership to Historic Lows (Bottom)



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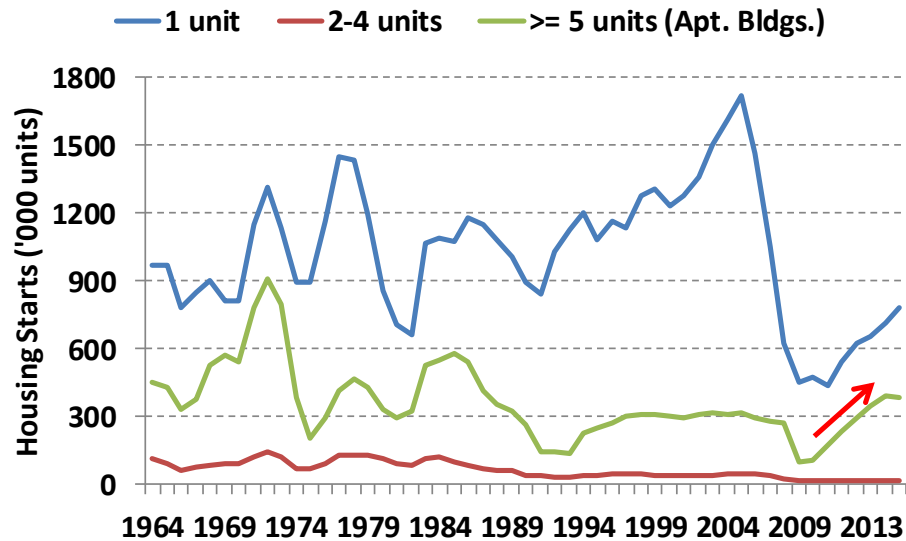


Source: Census Bureau, Citi.

Multifamily Property Appreciation Higher Than Single-Family

This has naturally led to faster growth in multifamily construction as well as higher multifamily appreciation rates.

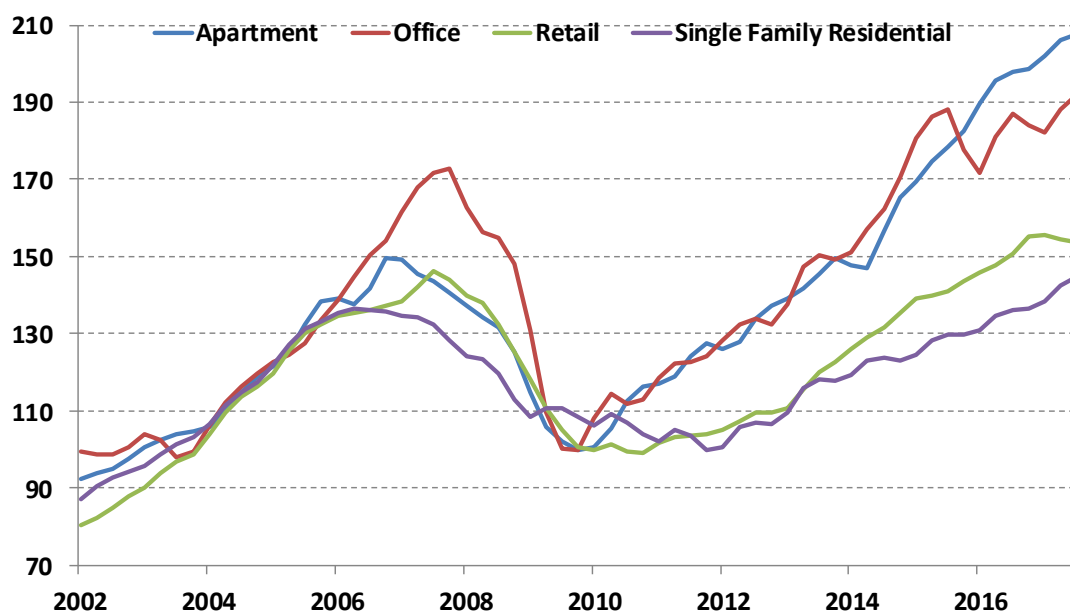
Figure 2. Apartment (5 units or more) Construction Reaches Highest Levels Since 1980s



Source: Census Bureau, Citi.

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Figure 3. Apartment Property Values Outperform Other Sectors During Economic Recovery



CoStar may revise their Apartment value index, resulting in even a greater degree of outperformance.
Source: CoStar Group, Case Shiller, Citi.

Healthcare Properties

While there is less data available for the healthcare property sector, anecdotal evidence suggests that it has been strong as well.¹

What Are FHA Project Loans?

In addition to its better known and much larger single family residential mortgage programs, FHA plays a substantial role in multifamily and health care. Meeting the need for affordable rental housing is in fact one of FHA's strategic goals.² Similar to its single family programs, FHA provides insurance on mortgages for multifamily

¹ "Experts Predict Another Good Year for Health Care Real Estate," Wall Street Journal – Developments," January 9, 2015. <http://blogs.wsj.com/developments/2015/01/09/experts-predict-another-good-year-for-health-care-real-estate/>

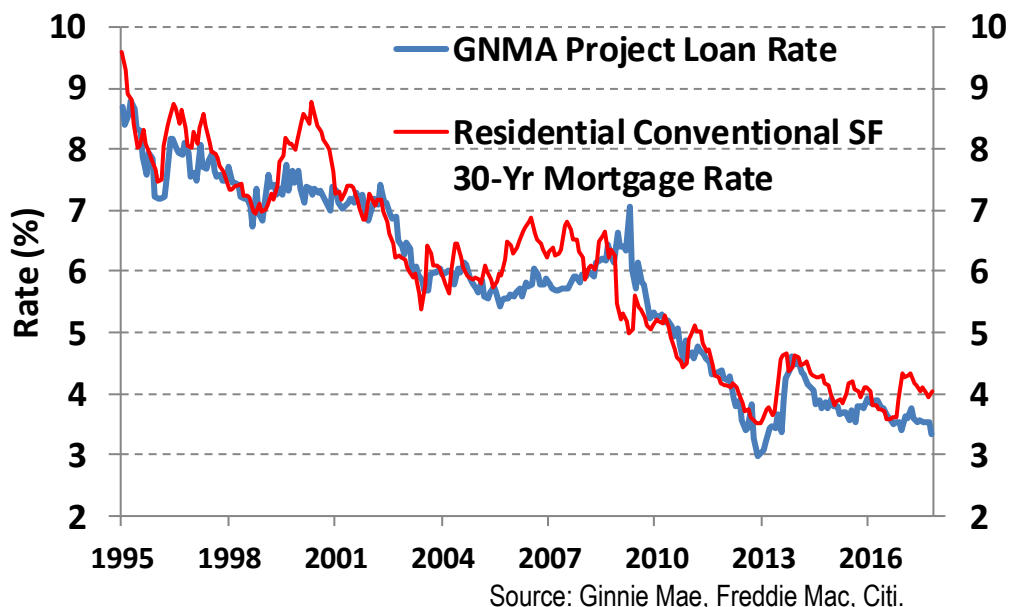
² "FHA Annual Management Report – Fiscal Year 2014," Federal Housing Administration, Department of Housing and Urban Development, November 17, 2014.

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and healthcare facilities.³ FHA's multifamily housing insured portfolio totaled 10,669 mortgages with \$69.5 billion outstanding, while FHA's healthcare facility portfolio totaled 3,125 loans with \$23.1 billion.⁴ (FHA single family totaled 7.8 million loans with around \$1.1 trillion outstanding.) Most of these multifamily and health care "project loans" are securitized into Ginnie Mae pools.⁵

Unlike the commoditized single-family residential mortgage market, the project loan sector is more specialized. For example, in contrast to widely advertised single-family mortgage rates, lenders typically do not publicize their FHA project loan rates - Figure 4 shows estimated project loan rates based on rates paid on originations of actual project loans.

Figure 4. Ginnie Mae Project Loan Rate Moves Similar to Single-Family Mortgage Rate



³ Like single-family FHA loans, FHA project loans are charged a Mortgage Insurance Premium (MIP).

⁴ As of Sep 30, 2014 – see "FHA Annual Management Report – Fiscal Year 2014," Federal Housing Administration, Department of Housing and Urban Development, November 17, 2014. This compares with 10,221 mortgages with \$57.2 billion outstanding multifamily and 2,957 loans with \$29 billion healthcare as of Sep 30, 2012 ("FHA Annual Management Report – Fiscal Year 2012," Federal Housing Administration, Department of Housing and Urban Development, November 15, 2012).

⁵ In addition to FHA, as discussed later, the US Department of Agriculture (USDA) Rural Development (RD) also contributes (a relatively small amount of) rural rental housing collateral that is securitized into Ginnie Mae "project loan" pools and REMIC deals.

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Advantages of FHA Financing

FHA project loan financing offers a number of advantages:

- 1) 35-year (40-year for some programs) financing
- 2) Smaller prepayment penalty
- 3) Has in the past offered higher LTVs than available from traditional funding sources⁶
- 4) Fairly flexible for new construction (for example, non-recourse)

Project Loan Program Overview

There are many Ginnie Mae securitized project loan programs⁷, but we can broadly classify them in Figure 5 covering the main project loan “Sections”.⁸

Residential Programs

221(d)(4) – insures mortgage loans to facilitate the new construction or substantial rehabilitation of multifamily rental or cooperative housing for moderate-income families, elderly, and the handicapped.

223(a)(7) – refinances multifamily rental properties that have FHA insured loans – no cash-out permitted (loan size limited by lesser of original size and current balance plus closing costs including prepay penalties and rehab costs)

223(f) – purchase or refinancing of existing apartment projects (“cash-out refinancing” permitted, provide proceeds are spent on improving original property)

⁶ Especially for health care although that advantage has somewhat diminished due to lower maximum LTVs

⁷ http://portal.hud.gov/hudportal/HUD?src=/program_offices/housing/mfh/progdesc

⁸ Named after “Sections” of the Housing Act of 1937 as amended.

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Figure 5. FHA Project Loan Section/Program Overview Matrix

	Residential	Healthcare¹
Construction/Renovation	207, 221(d)(3) (non-profit), 221(d)(4) (for-profit)	232
Purchase and Refinance with Cash-Out Option (no cash-out for healthcare)	223(f) if not already FHA insured; If already FHA insured, sometimes another section in combination with 223(f)²	232/223(f)
Streamlined Refinance allowing (a) draw back to orig loan amt and (b) term reset (no re-appraisal)	223(a)(7) - Usually in combination with another section above	232/223(a)(7)
“Streamlined Refi” Loan Modification – only rate change with everything else staying the same³	LM Pool Type³	LM Pool Type³
Second/Supplemental	241⁴	241⁴

¹ In addition to Section 232, which serves Residential Care Facilities, Section 242 serves Hospital Facilities – but since they are not in REMICs (see Property Type Effect – Residential vs. Healthcare), we place more emphasis on Section 232.

² For example, 207/223(f) (but does not seem to apply to 221(d)(4)).

³ Unlike loan modifications applied to delinquent loans, the “LM” does not imply any loan distress. LM is actually a Ginnie Mae pool code, but we include it here because it acts like a special streamlined refinancing “Section”.

⁴ Section 241 often shows up as 241(a) or 241(f).

Source: HUD, Citi.

Healthcare Programs

232 – purchase, refinance, new construction, or substantial rehabilitation of nursing homes, assisted living facilities, and board and care facilities (a combination of these uses is acceptable - e.g. refinance of a nursing home coupled with new construction of an assisted living facility)

242 – purchase, refinance, expansion, modernization, or remodeling of acute care hospital facilities ranging from large teaching institutions to small rural critical access hospitals

USDA Rural Rental Housing

There is also a small but somewhat growing set of rural housing Ginnie Mae pools.

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They are insured by the U.S. Department of Agriculture's (USDA) RD program under Section 538/515 that provides a 90% guarantee (Ginnie Mae provides another layer of guarantee for the pooled loans).

LM Pool Type – Streamlined Refinance Loan Modification

A project loan in a Ginnie Mae LM pool is one that has been rate-modified with FHA approval after final endorsement. The previous pool is prepaid and the new LM pool type is issued, but loan maturity and balance stay the same. "Modification" here is not an indication of any duress of the original loan.

LS Pools – Multiple Loans in a Pool

While the bulk of the pools are one loan-one pool structures, there is an LS pool code that allows securitization of multiple small loans into the same pool. Much of USDA production are in LS pools due small loan sizes – also note that GNMA does not allow mixing of HUD and USDA loans in the same LS pool.

RX Pools – Mark-to-Market Loan Modification

Here the basis of the previous mortgage is reduced to the market level (essentially a principal reduction loan modification), previous loan paid off, and the shortfall amount is carried as a second/third mortgage on which payments are only made to the extent funds are available.

Collateral Characteristics

As discussed earlier, the majority of project loans are multifamily residential. However, even after excluding very large Section 242 Hospital loans (about \$80 million per loan on average), the average health care project loan size tends to be larger than that of multifamily housing.

In general, loan sizes can vary quite a bit, ranging from around \$1 million to \$100 million. And, geographically, FHA projects are distributed across the country as one might expect – states with larger populations possess more loans, and the average size of a loan roughly correlates with per-capita GDP. However, there are higher than average concentrations (relative to state size) in certain New England/Mid-

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Atlantic states – this is especially the case for the District of Columbia.⁹

Construction vs. Property Management

There are two flavors of project loans – Construction Loan Certificates (CLC) and Permanent Loan Certificates (PLC).

CLCs are, as the name implies, intended for construction. An overall loan amount is approved and during the construction period there may be multiple drawdowns that total up to the approved loan amount. During construction, interest is only paid on the amount drawn down and no principal is repaid. Upon the completion of construction and subsequent HUD approval, CLCs are converted to PLCs. There is usually a specified conversion date. Penalties may apply if the conversion date is extended.¹⁰ These penalties are not guaranteed by GNMA.¹¹

PLCs are loans for the purchase, rehabilitation and operation of existing properties. They may be converted CLCs or loans that originated as PLCs. They are all fully amortizing.

We have seen that CLCs are issued at higher WACs than PLCs of the same vintage, by 50 bps on average. We believe that this is mainly due to the hedging costs. The purchase of a CLC is a commitment to a certain notional amount, only a portion of which may be drawn at the time of origination. The rest of the amount will be drawn at various points in the future that are uncertain. To protect against interest rate risk the undrawn amount must be hedged. To compensate CLC buyers for this cost the borrower must pay a higher rate. Note that the notional of the hedge is reduced with each successive draw as seen in Figure 6.

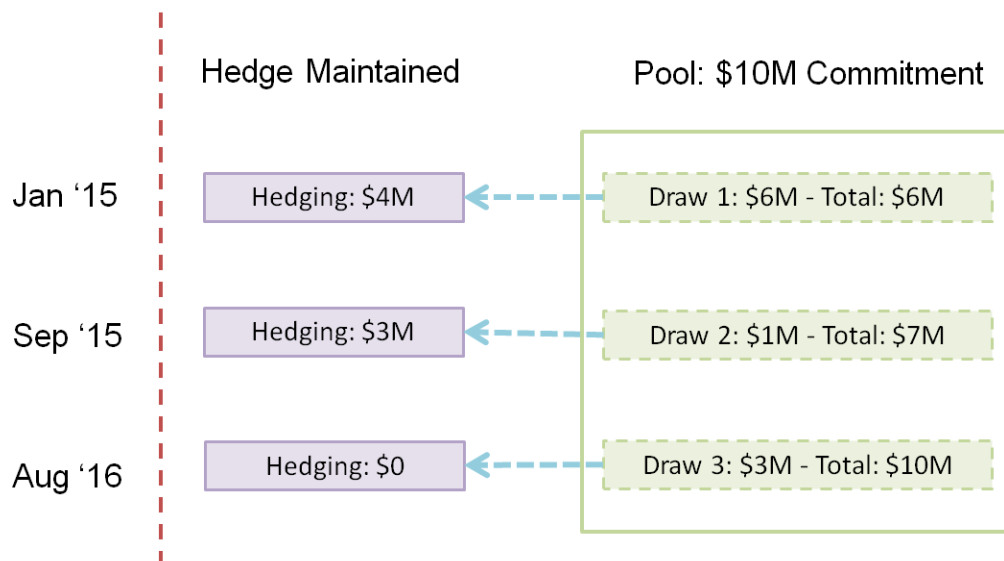
⁹ Approximately 81 percent (around 11,000 loans total) of all current project loans are in metropolitan (as defined by the U.S. Census) areas in line with the distribution of the general population; the rest are in micropolitan (around 1,600), small town (around 100), or rural (300) areas (Rural Health Research Center).

¹⁰ For examples of conversion (and drawdown) penalties please contact the authors of this paper.

¹¹ These “penalties” should not be confused with the more common prepayment penalties discussed in the next section.

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Figure 6. CLC Draws vs. Hedging of Undrawn Commitment Example



Source: Citi.

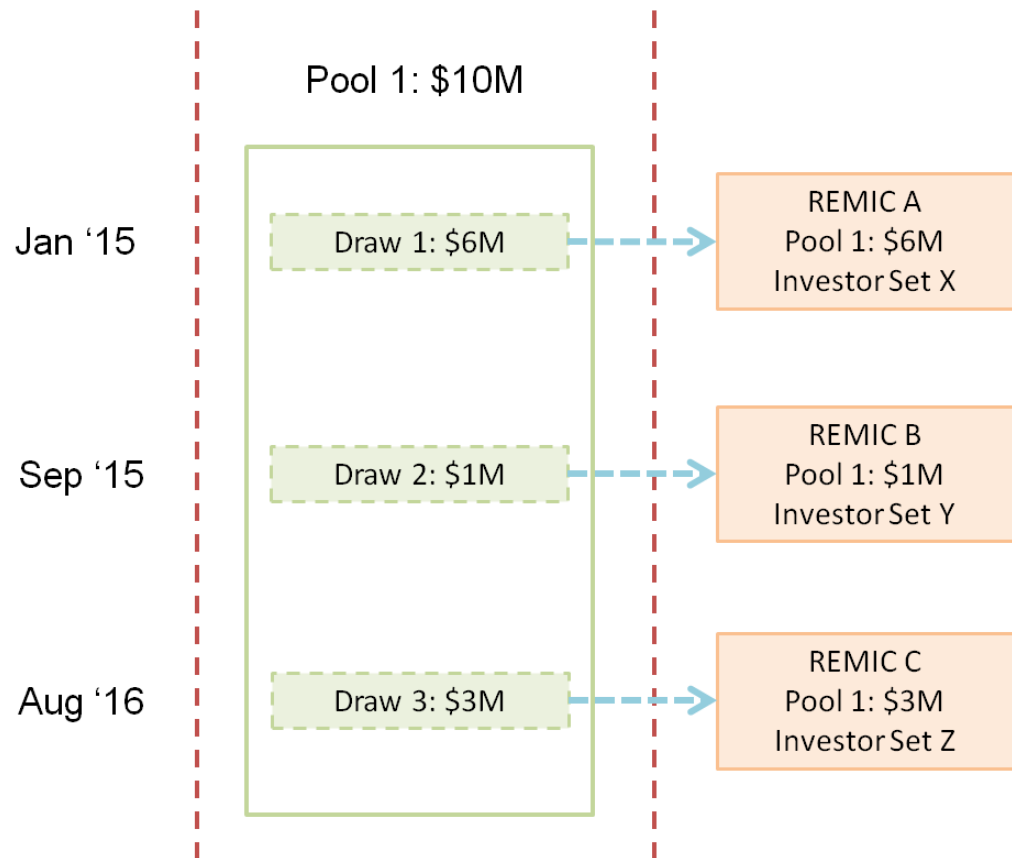
Construction Loan Securitization

In general originators sell pools to broker/dealers. The broker/dealers in turn securitize pools into REMICs and sell the REMIC tranches to end investors. Construction loan securitization requires further explanation.

As stated in the previous section when a broker/dealer buys a CLC pool, the broker/dealer is committing to the full loan amount. However, the initial draw amount is usually some fraction of that loan amount. In this situation the broker/dealer securitizes only the drawn portion of the CLC into a REMIC.¹² When the next draw takes place that portion of the CLC is then securitized into a new REMIC and so on for each additional draw. There is no requirement for an investor to participate in each of the REMICs. To reiterate it is only the broker/dealer and not the end investor who is committed for the full loan amount of the CLC. That is why the Investor Sets are different across the REMICs in Figure 7.

¹² Broker/dealers keep project loans on their balance sheet from the time of purchase to securitization and sale of the REMIC, which can take a month or two. For this time they hedge the interest rate risk from the fixed rate earning pools vs. the floating rate funding that they have to pay. Note that the notional being hedged is the full pool amount in the case of the PLC or the drawn but as-yet-un-securitized portion of the CLC.

Figure 7. CLC Securitization into REMICs



Source: Citi.

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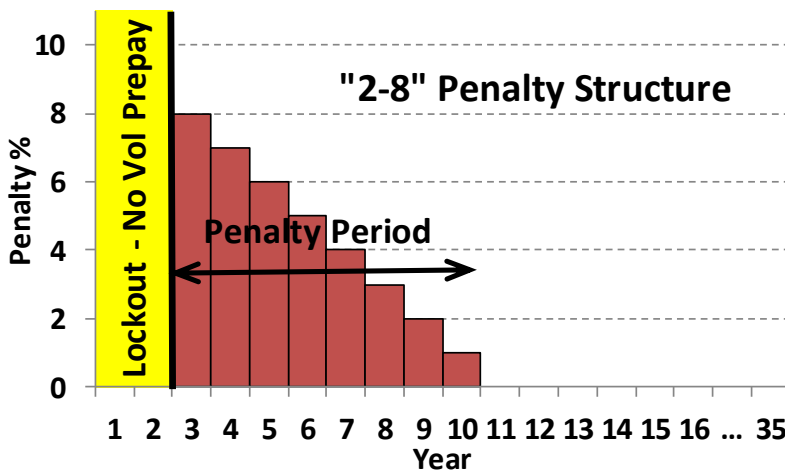
Primary Prepayment Drivers

Restrictions on Prepayment

Most project loans have the following prepayment restrictions:

- 1) **Lockout Period** – Most project loans start with a period during which voluntary prepayments are not allowed. The length of an initial lockout period can vary greatly – from as little as one month to several years (some project loans have 10-year lockouts). Longer lockout periods were more common a decade ago – shorter lockout periods have become more common in recent years.¹³
- 2) **Prepayment Penalty Period** – The other common feature of project loans is a prepayment penalty period, which follows the end of any lockout period that exists. For example, an initial 24-month lockout period might be followed by an 8-year penalty period for which the penalty amount starts at 8% of the prepaid amount and declines by 1% per year.¹⁴

Figure 8. Lockout and Prepayment Penalty Illustration



Source: Citi.

¹³ FHA has the authority to override lockout and prepayment terms if it determines that it is in the best interest of the federal government. Additionally 223(f) loans cannot be prepaid for 5 years without written approval from the FHA, regardless of the loan terms.

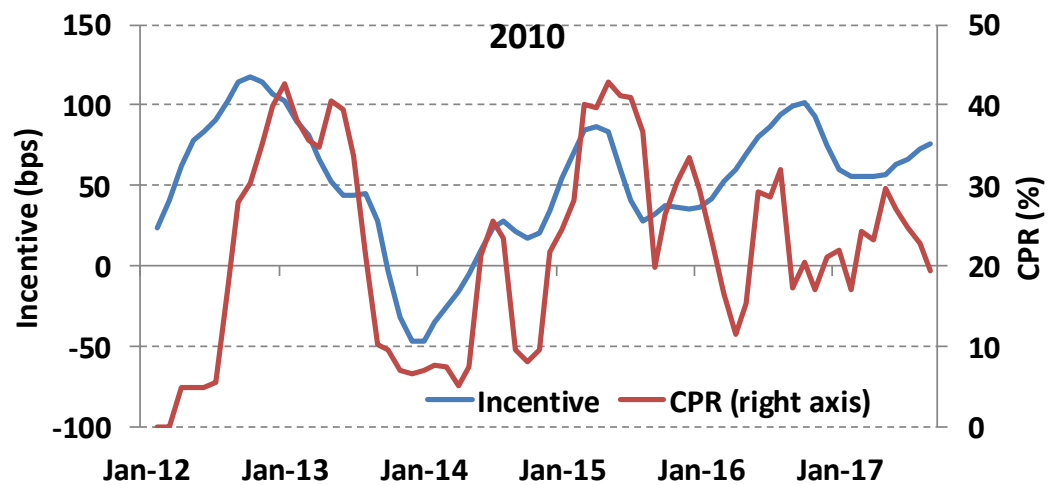
¹⁴ More precisely from an offering circular supplement: “Lockout up to but not including the Lockout End Date; thereafter a Prepayment Penalty of 8% of the prepaid amount up to but not including the twelfth mortgage loan payment date beyond the Lockout End Date disclosed”, “declining thereafter by 1% annually up to but not including the Prepayment Penalty End Date.”

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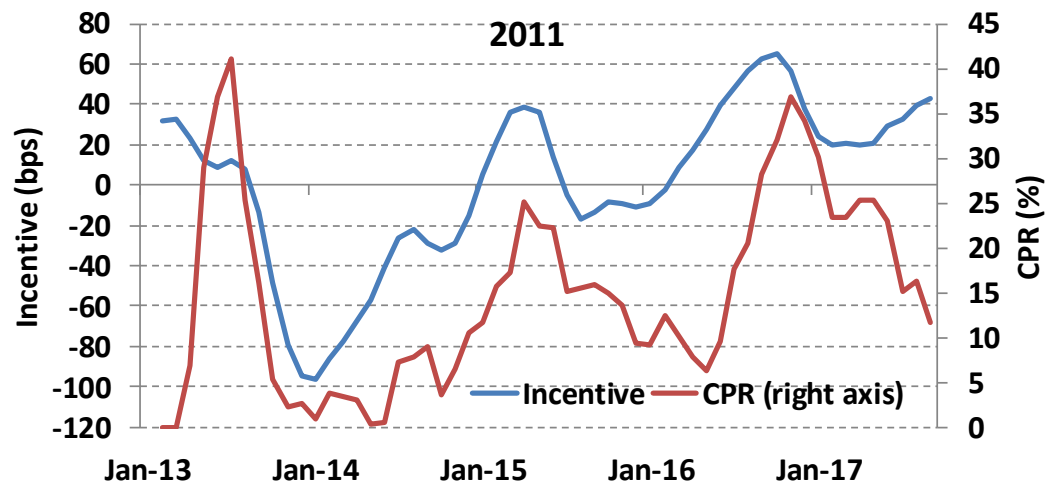
Rate Driven Refinancing

Refinancing Incentive Impact – The data shows that the size of the refinancing incentive matters. Figure 9 considers 2010 (top) and 2011 (bottom) originated collateral. In both cases an increasing rate incentive leads to a subsequent increase in the CPR. The decreasing rate incentive, in the latter half of 2012 and early 2013, leads to a subsequent decrease in CPR.

Figure 9. Rate Incentive compared to CPRs for 2010 and 2011 originations



2010 originated collateral with ~2 year lockout period and WAC 4.53%.



2011 originated collateral with ~2 year lockout period and WAC 4.10%.

Note Incentives and CPRs are 3-month moving averages.

Source: Ginnie Mae, Citi.

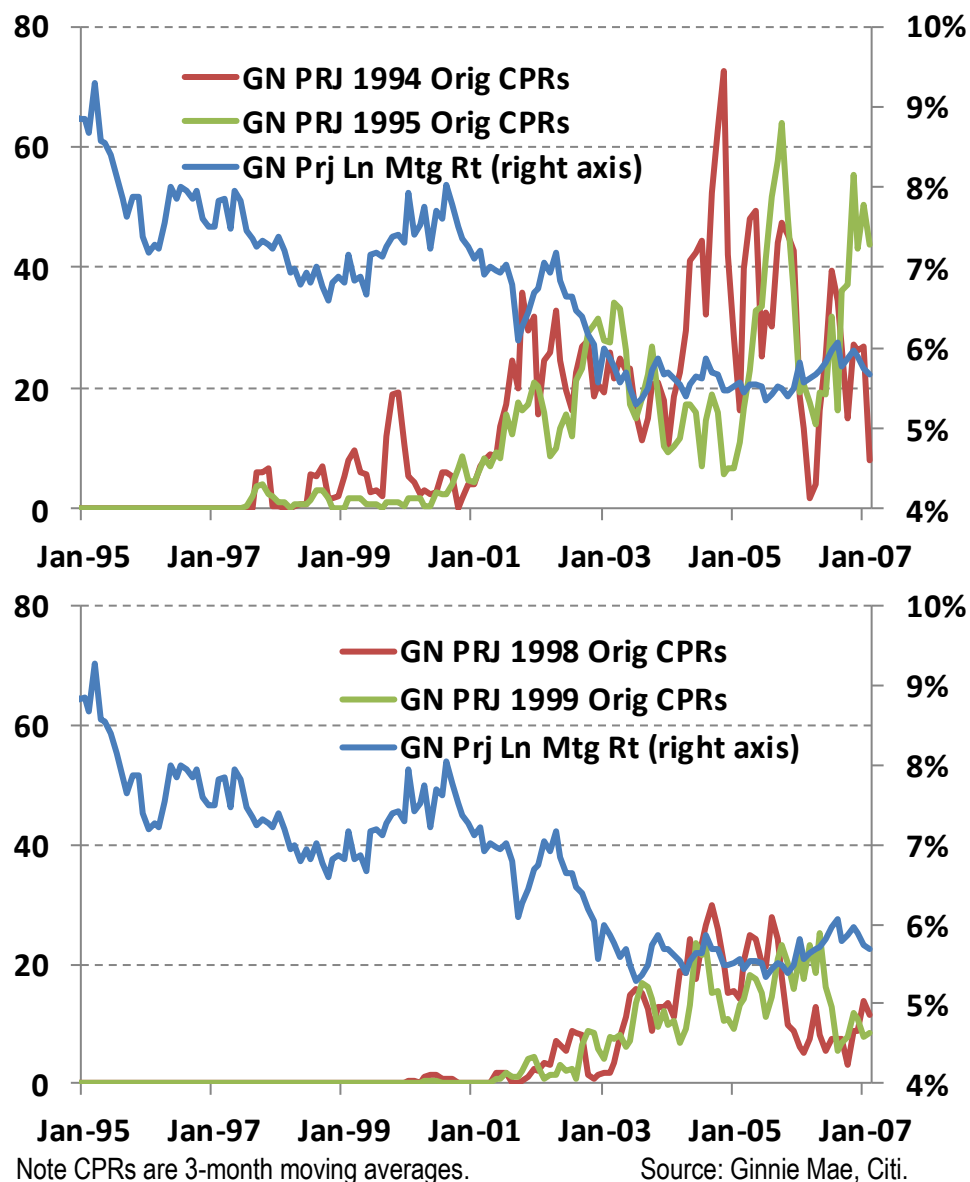
Figure 9 also indicates that there may be a lag between the appearance of a rate

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incentive and its translation into a refinancing. Industry participants have suggested that the refinancing process is lengthy and in some cases can take up to 6 months. However, as discussed later, recent data suggests a much shorter lag.

Further evidence of the impact of rates can also be seen in the run-up in speeds in 2001-03 when rates rallied approximately 200 bps. Figure 10 top panel shows 1994-95 origination prepayment speeds. For later 1998-99 originations, average 6 to 7 year lock-outs probably suppressed prepayments until the bubble years (Figure 10 bottom panel).

Figure 10. Speeds High During 2001-03 Rally and 2004-06 Property Bubble – 1994-95 Orig. (Top), 1998-99 Orig. (Bottom)



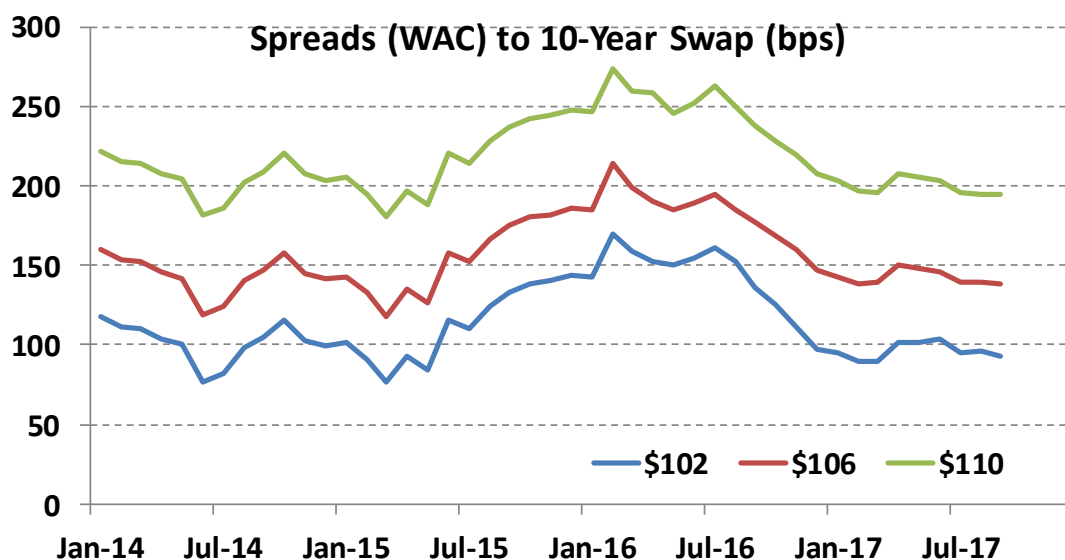
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Project Loan Rate

Projecting the refinancing rate requires a view on the prevailing Project Loan Rate. However, there is no market observed “Project Loan Rate”. Unlike the single family world there are no TBAs to serve as a reference. Also, the rate that is charged is impacted by prepayment assumptions and the penalty, if any, to be received in such instances. Refinancing borrowers prefer no-cost refinancing i.e. using the proceeds from the new loan to cover the prepayment penalty due on the old loan. Hence, such pools are traded above par with rates to match.

A reasonable assumption for the project loan rate is to add a spread to an observable reference rate. In practice, dealers quote multiple price/rate¹⁵ combinations to account for no-cost refinancing described above. Figure 11 shows the spreads, to the 10-year Swap, for a number of price points.

Figure 11. GNMA Project Loan Rate spreads to the 10-Year Swap Rate by Dollar Price



Source: Citi.

Note that this approach results in a penalty adjusted project loan rate. There are 5 of them as rates for the \$102, \$104, \$106, \$108 and \$110 prices are received. As a

¹⁵ Dealers quote coupon, not WAC. However, the guarantee and servicing spread is 25 bps for most pools (almost 75% of those issued in 2017 YTD). For small size or bespoke pools the spread may be different. Hence, one can assume that WAC = coupon + 25 bps for projection purposes.

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rule of thumb, X points above par corresponds to a penalty of $X - 1$ penalty points. So the 5 rates above are assumed to be for the 1%, 3%, 5%, 7% and 9% penalty points. Linear interpolation and extrapolation is used to get the rates for the 0%, 2%, 4%, 6%, 8% and 10% penalty points. This approach allows the model to treat each pool according to the refinancing penalty it faces at any given time, hence, the use of the term penalty adjusted project loan rate.

A second advantage of this approach is that the model reacts to daily changes in the market as project loan rates are received every day. The \$102 i.e. the 1% penalty, rate is received daily and converted to a spread resulting in a series of spreads for the 1% penalty. The current spread for the 1% penalty is the average of the spreads received for the 1% penalty for the month. For projection purposes this current spread is kept constant and just added to the projected 10-year swap rate to get the 1% penalty project loan projection. Similarly the projected project loan rates for the other penalties are projected.

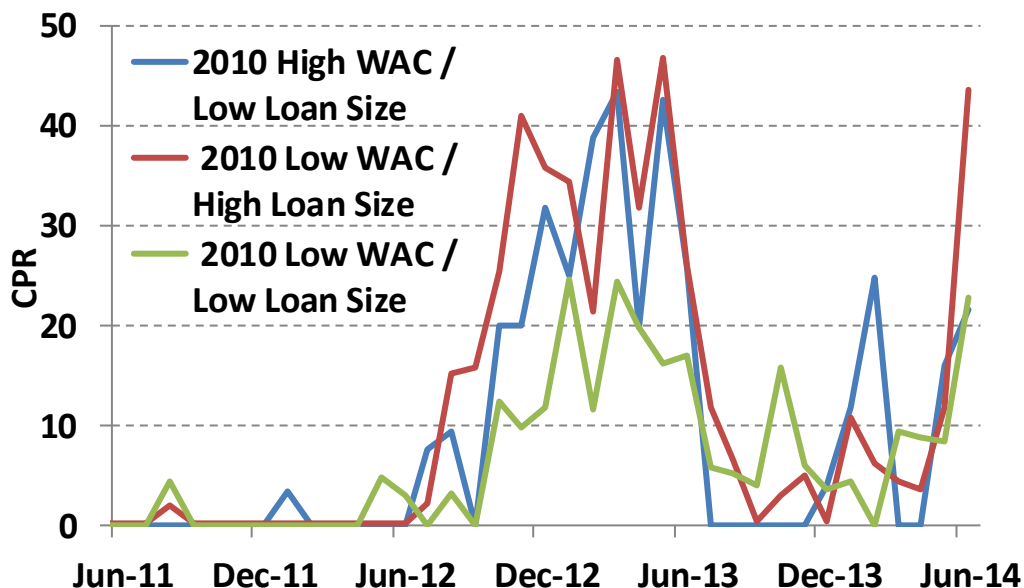
Keep in mind that project loans lock in a rate some time before the loan is closed. In the case of LM refinancing this can be as little as 5-7 weeks before closing. In the case of new construction, 221(d)(4), loans it can sometimes take 6 months. Hence, when projecting a prepayment in a given month one needs to look at the project loan rate from a couple of months ago. To account for the variance in rate locking periods the model uses a blended rate over the last few months instead of any one month.

Loan Size Impact on Refinancing Incentive

As is the case for single family residential, large project loans tend to prepay faster than smaller project loans (Figure 12); in general, doubling the loan size doubles the refinancing incentive/savings.

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Figure 12. Higher Loan Size Results In Faster Speeds Just As Higher WAC Results in Faster Speeds



High WAC / Low Loan Size is 5.13 / \$3.2 M.
 Low WAC / High Loan Size is 4.25 / \$15.3 M.
 Low WAC / Low Loan Size is 4.32 / \$3.8 M.

Source: Ginnie Mae, Citi.

Prepayment Penalty

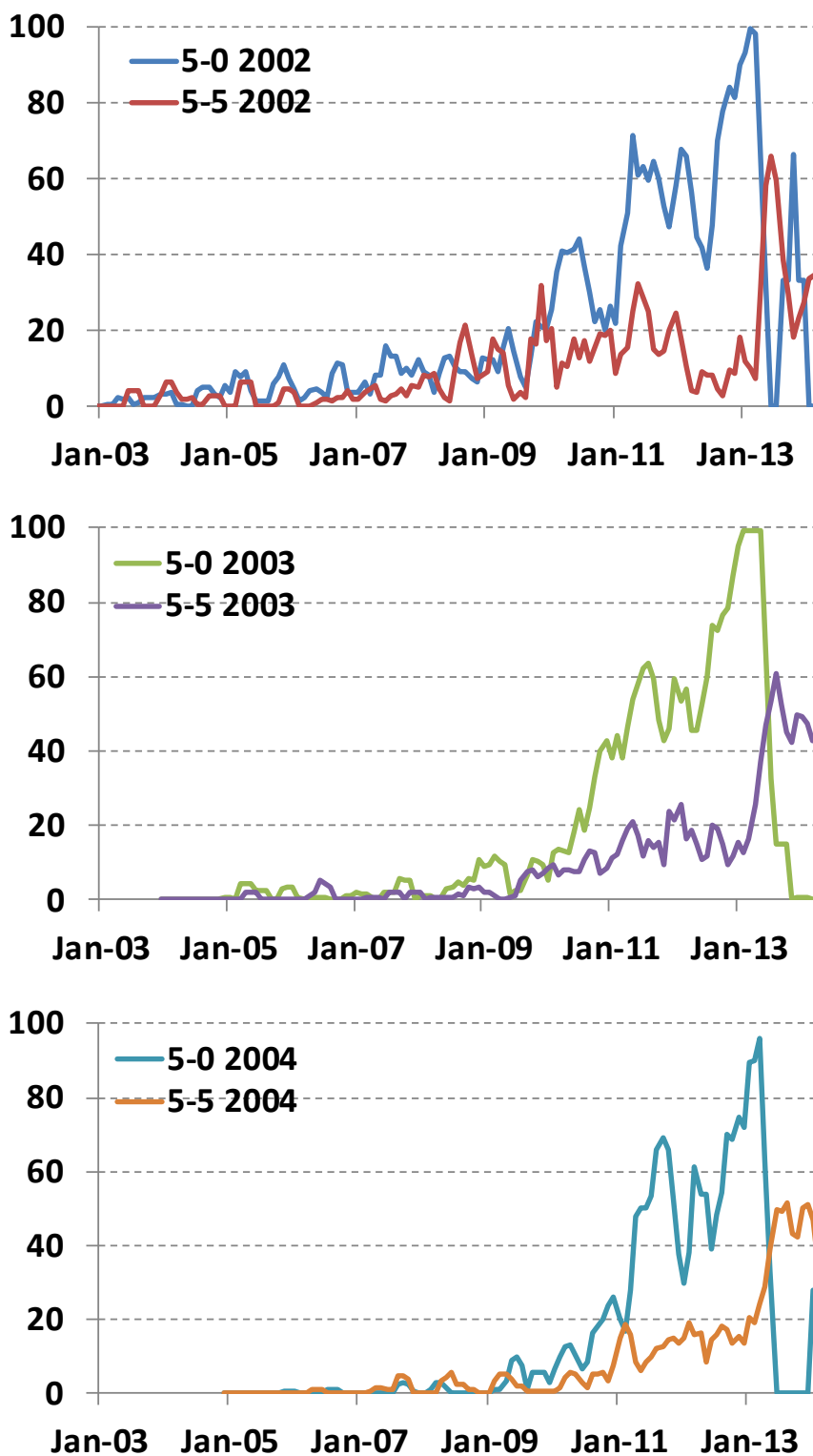
While the lockout prevents prepayment during the initial period of the loan, the prepayment penalty suppresses it. The impact of the prepayment penalty can be observed by comparing cohorts of loans originated in 2002-2004. In Figure 13 each cohort has a 5 year lockout period. However, for each year the cohort of loans with no prepayment penalty was compared against that with a 5 year prepayment penalty, 5-0 vs. 5-5.

In each case the 5-0 cohorts all prepaid faster and almost completely. Conversely the 5-5 cohorts paid off slower and at a more or less constant speed, up until the final year of the prepayment penalty period, when the penalty was presumably quite low.

Note that Figure 13 is an extreme example of the impact of prepay penalty. More recently, the impact of the penalty has been muted, suggesting that there might be hidden collateral differences in the cohorts shown in addition to the difference in penalty.

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Figure 13. Prepay Penalty Effect for 2002-04 Cohorts with and without Prepay Penalty Periods



WAC: 2002 – 6.6%; 2003 – 5.7%; 2004 – 5.7%. Source: Ginnie Mae, Citi.

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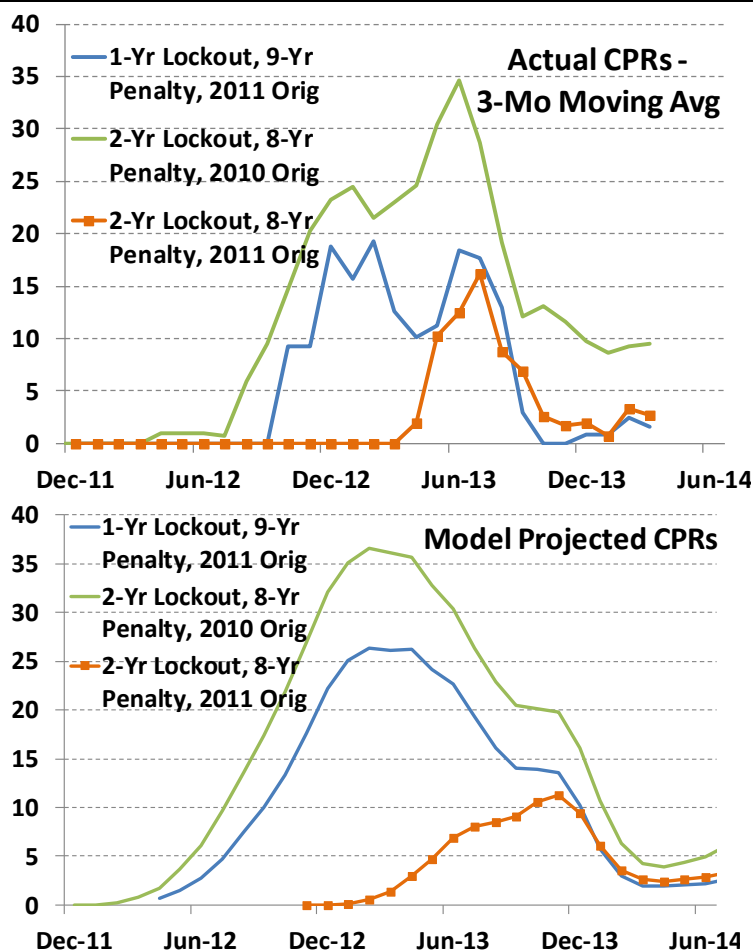
More Voluntary Prepay Drivers

Lockout and Pent-Up Demand

One of the prominent patterns of project loan voluntary prepayments is the jump that often occurs after the expiration of the lock-out period and in the initial few years of the prepayment penalty period.

Figure 14 shows how speeds on three cohorts jumped as the lockout period on each expired.

Figure 14. Speeds Jump After Lockout – Larger Rise on Higher WAC 2010 Loans – Actuals (Top) and Model Proj (Bottom)^a



^a Approximate 1-yr lockout on 2011 originated collateral (WAC 4.14%) and 2-yr lockout on 2010 originated collateral (WAC 4.61%) expires in 2012 when speeds can be observed to jump while 2-yr lockout on 2011 originated collateral (WAC 4.13%) jumps around a year later when its lockout expires. 3-month moving average CPRs are shown in both charts.

Source: Ginnie Mae, Citi.

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The longer the lockout, the greater is the release of pent-up demand at its expiry. Thus, a loan with a 2 year lockout would naturally have more pent-up demand than one with a 2 month lockout. Additionally prepayments naturally season as the prepayment process is complex and often can't start till a few months before the lockout expiry. Thus, the jump from an expiring lockout is really a combination of the length of the lockout, the pent-up demand and the seasoning.

While there can be little doubt that the lockout expiration has an impact on project loan speeds, a natural question is why speeds in Figure 14 jumped despite large prepayment penalties (probably typically 8% to 9% of the prepaid amount) that were in effect upon the expiration of the lockout period.

We believe there are a few possible explanations:

- a) **Record low rates** – some borrowers had substantial incentive to refinance into record low rates that reached around 3% (see Figure 4) – an 8% to 9% penalty was paid immediately in order to secure up to 30+ years of around 1% interest rate savings per year (if say, the rate declined from 4% to 3%)
- b) **Property appreciation an extra incentive** – some borrowers probably also had accumulated property appreciation that they have wanted to monetize
- c) **Penalty tax deductible** – some of the sting of the penalty may be reduced by favorable tax treatment; the penalty can be deducted as interest in the year incurred, reducing the project manager's tax liability¹⁶

Most issuance today has no lockout period. As such lockout expiration has little effect in projections for recently issued collateral.

IRR/LM Refinancing

The spate of high penalty terminations lockout expiry from late 2014 to the middle of 2015 and to a lesser extent in the middle of 2016, shown in Figure 15, supports the view that property managers have become more astute over the last few years.

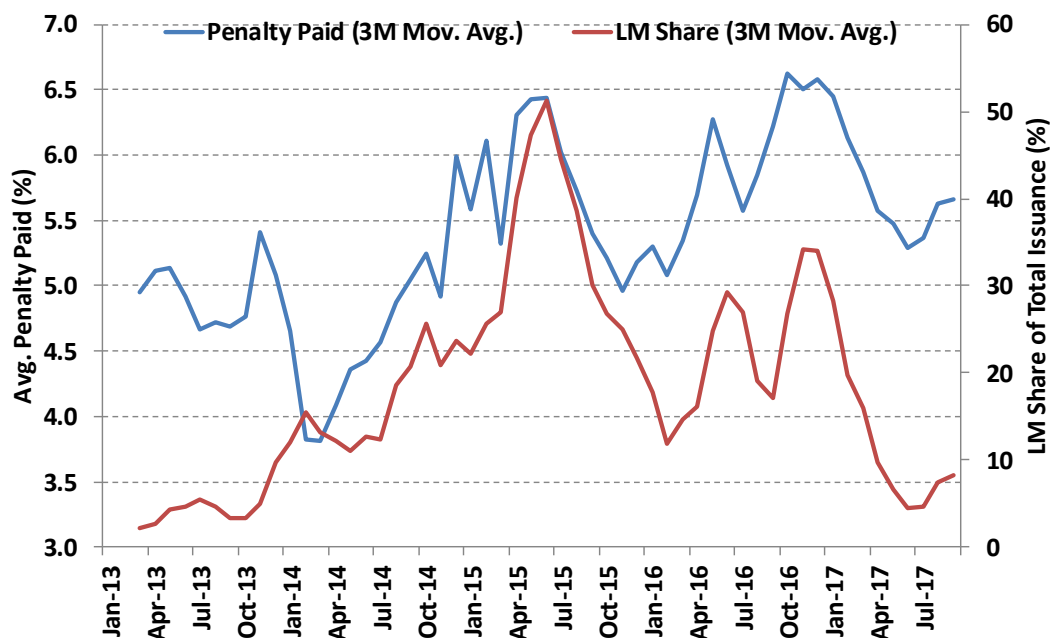
Specifically, they seemed to have been taking advantage of a streamlined form of refinancing where only the rate is changed and all other terms remain the same. This results in a shorter refinancing process; it takes roughly 2 months as opposed

¹⁶ See IRS Publication 535, <http://www.irs.gov/publications/p535/ch04.html>, and confer with a tax advisor.

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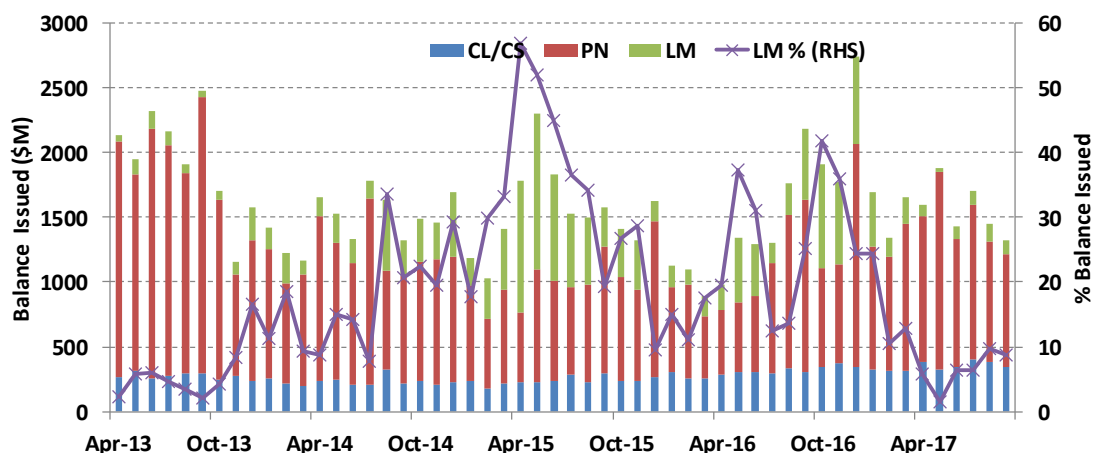
to the usual 6. While the new loan retains the same FHA Case Number, it is securitized into a new GNMA pool with the pool code LM, for Loan Modification.

Figure 15. Prepayment Penalty Paid vs. LM Share of GNMA Project Loan Issuance



Source: Ginnie Mae, Citi.

Figure 16. LM Issuance as a share of total GNMA Project Loan Issuance



Source: Ginnie Mae, Citi.

Figure 16 shows that since mid-2013, LM pools become a large proportion of the total GNMA Project Loan universe during refinancing waves.

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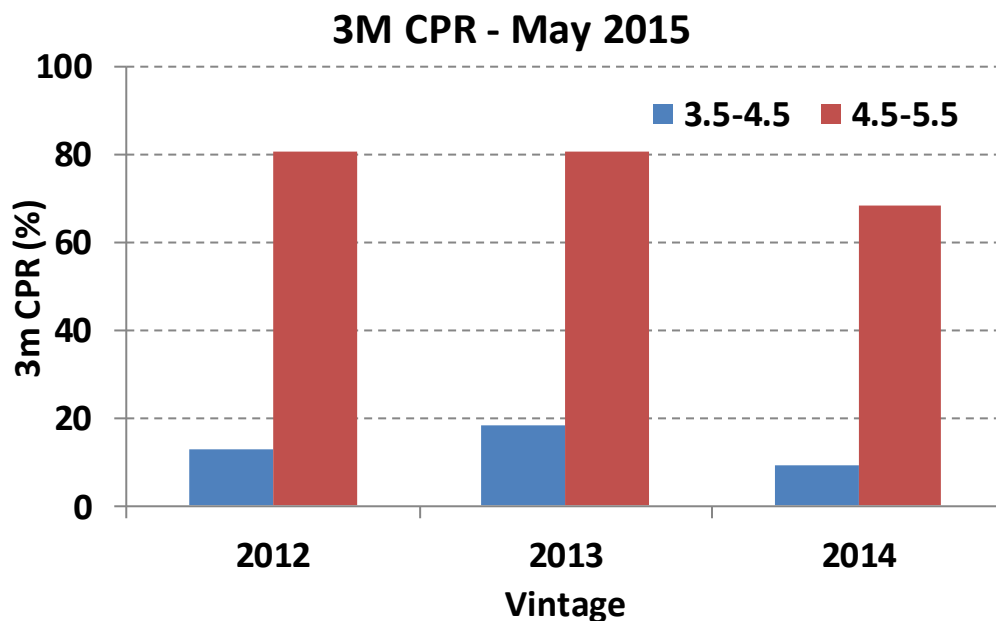
In the current pricing environment borrowers are able to get “no cost” refinancing, even though they are paying the prepayment penalty on the original loan. In effect the cost of the penalty is being absorbed by the investor in the new LM pool. In exchange the borrower pays a higher rate than he would have been able to get without a penalty, but still lower than his current note rate.

A good analogy, in the residential mortgage world, is a borrower refinancing his current loan to a lower rate, but saving on closing costs by receiving negative points in exchange for not getting the lowest rate possible.

Our conversation with industry participants, confirm that in late 2017 newly issued LM pools traded in the 108-110 price range with a 4% coupon. For a similar price, the investor would have received a GNMA TBA with a 5% coupon. The difference is that the project loan prepayment can trigger a prepayment penalty. In essence, the project loan investor accepts a lower coupon in exchange for potential profit from the exercise of the prepayment option.

The results were some of the fastest speeds seen for Project Loans (Figure 17), even surpassing aggregate single-family speeds for the first time in a refinancing environment.

Figure 17. Fast Project Loan Refinancing Speeds in Early 2015



Source: Ginnie Mae, Citi.

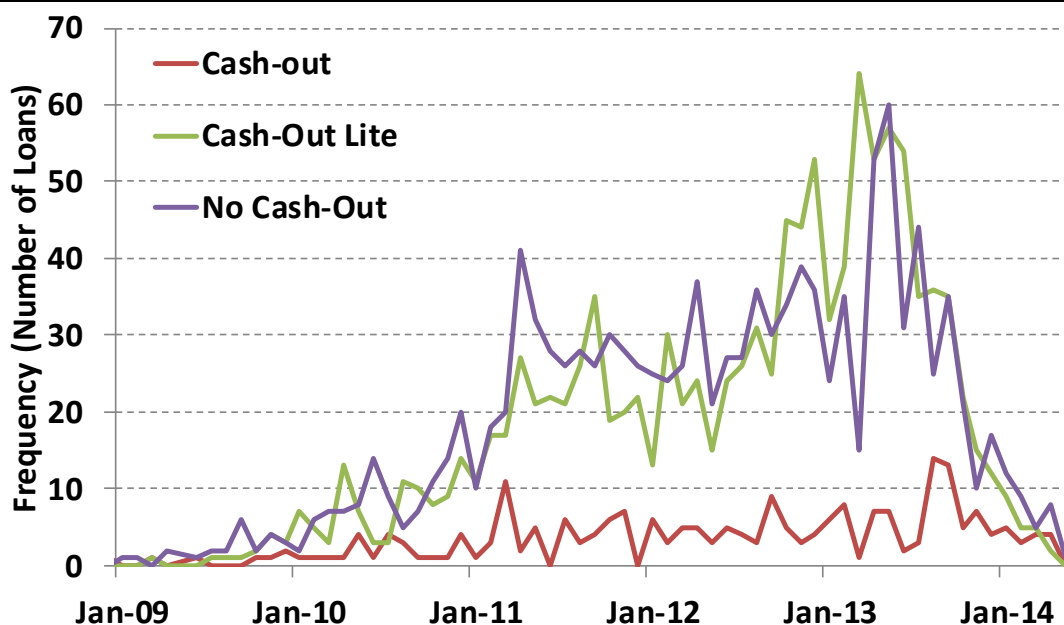
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Property Appreciation and “Cash-Outs”

While there may be few “pure” FHA-to-FHA cash-outs (for which there is no rate incentive),¹⁷ there are many cases of project loans refinancing back to the original balance of the loan. While such refinances would not provide much cash early in the life of the loan when little principal has been repaid, significant cash can be obtained by refinancing very seasoned collateral. Specifically, Section 223(a)(7) allows refinancing with the original mortgage balance being a cap on the new loan amount. One reason for not trying to exceed the original balance could be the additional hassle involved in trying to cash-out refinance beyond the original balance (it might take 6 to 9 months to do a “beyond original balance cash-out refinance” vs. 3 to 6 months for a Section 223(a)(7) original balance limited cash-out). Another advantage of “(a)(7)” is that there is no requirement for a re-appraisal.

Figure 18 in fact suggests that cash-outs with the loan amount limited by the original balance are quite common (as common as pure FHA-to-FHA rate refinances).

Figure 18. “Cash-Out-Lite” (Cash-Outs Not Exceeding Original Balance) Frequency Comparable to Pure Rate Refinances



Estimated FHA Project Loan to FHA Project Loan prepayment frequency categorized by amount of cash taken out. Some project loans may have cash-out refinanced into a conventional loan.

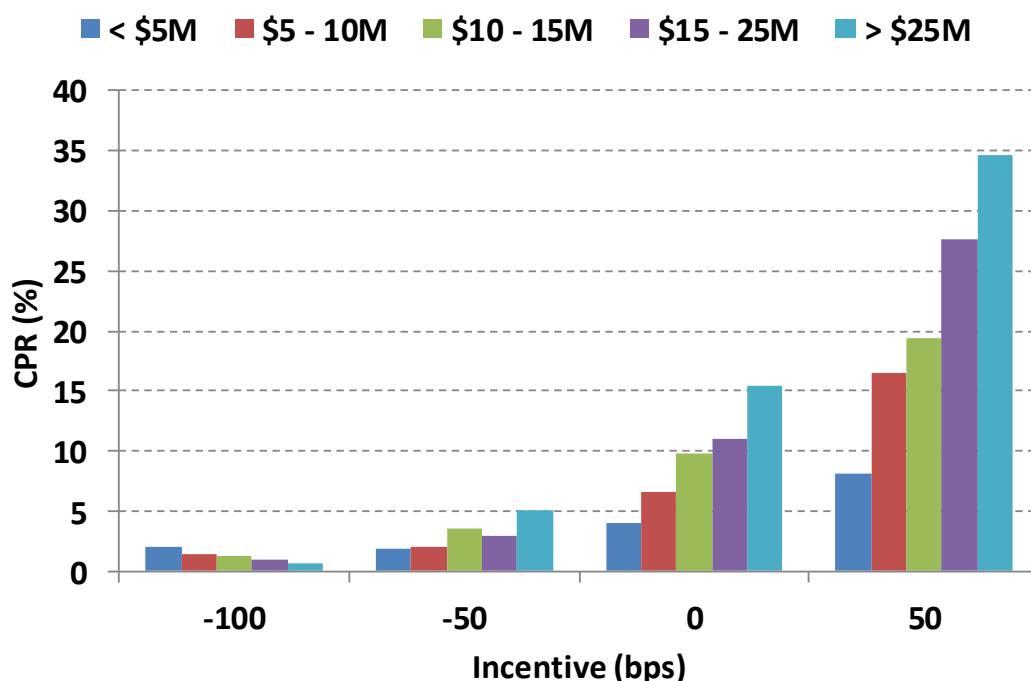
Source: Ginnie Mae, Citi.

¹⁷ Section 223(f) allows cash-out refinancing up to 85% of HUD’s appraisal value.

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One would expect out of the money loans to show some equity extraction, even in times of high rates. However, in two separate higher rate periods – 2000 (following 1993-4) and 2014 (following 2013) – few instances of such behavior were observed in our data. In particular, 1993-4 vintage loans prepaid at 1.5% in the one-year period from Q2 2000 to Q1 2001, during which these pools were out of the money. This is in line with longer-term out of the money prepayment speeds as seen in Figure 19.

Figure 19. Historical Out-of-the-Money Speeds and Just In-the-Money Speeds by Loan Size



Source: Ginnie Mae, Citi.

The bottom line is that most cash-outs are probably mainly driven by rate incentive (similar to residential cash-out refinances).

Media Effect

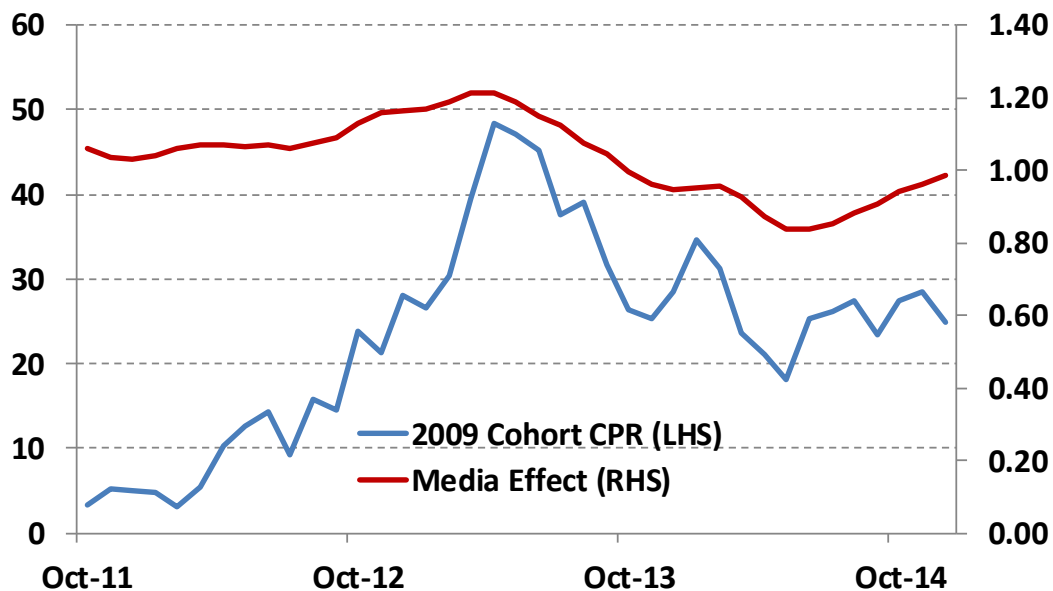
When rates are consistently falling one would expect prepayment speeds to be faster than that just implied by the incentive.

The 2009 vintage cohort experienced rising rates from inception to the end of 2012. The corresponding media effect was most pronounced from Q2 2011 to Q3 2012. Factoring in the 6 month lag to account for the time to refinance we would expect to

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see faster prepayment speeds from Q4 2011 to Q1 2013. Figure 20 shows a steep rise in prepayment speeds for the 2009 vintage cohort over that period.

Figure 20. Media Effect for the 2009 Cohort



The 2009 cohort has a WAC of 5.65%. CPRs and Media Effect are 3-month moving averages.
Source: Ginnie Mae, Citi.

Original PLC vs. Converted PLC

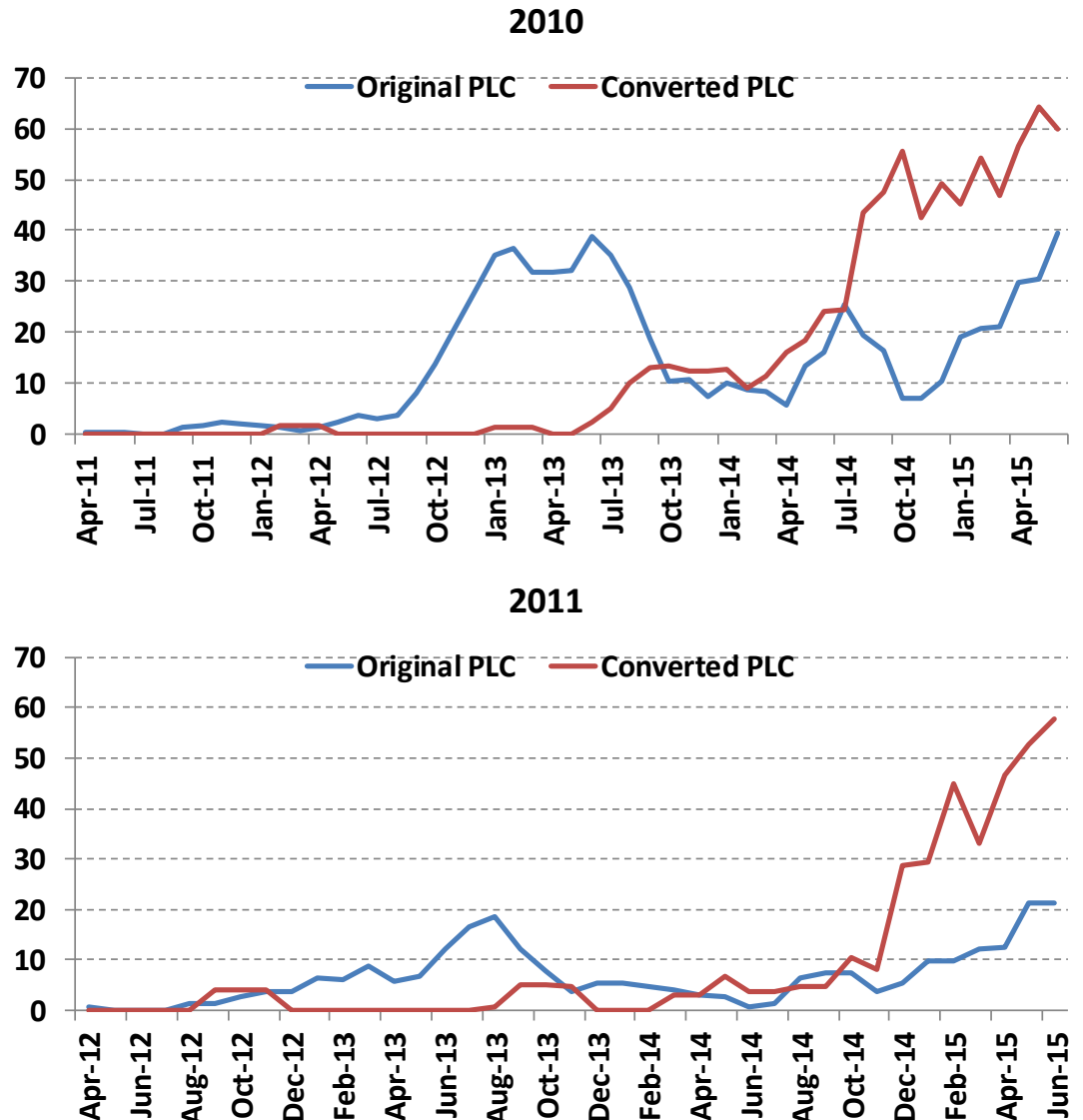
A question often asked by clients, is whether speeds on project loans that started out as PLCs (“Original”) are any different than speeds on those project loans that started out as CLCs and then converted to PLCs (“Converted”).

From Figure 21, it appears that converted PLCs pay faster. But for a given origination year, the converted PLCs have higher WACs, as referenced earlier in *Introduction* → *Project Loan Program Overview*. The faster speeds initially seemed to be a consequence of converted PLCs simply having a greater rate incentive.

However, an analysis of the model projections still showed a residual difference in speeds and necessitated the introduction of a factor. While the reasons for this difference are not completely understood, it is surmised that the converted CLCs pay faster due to the presence of operators who primarily construct to sell. Once the property has been built and uncertainty over the project’s viability has been removed, the property can be sold at a higher valuation.

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Figure 21. Original vs. Converted PLC Speeds (3-month averages) for 2010-11 Vintages



2010 WACs: Original PLCs 4.62% vs. Converted PLCs 5.45% → 83 bps difference

2011 WACs: Original PLCs 4.15% vs. Converted PLCs 4.86% → 71 bps difference

The upper chart shows actual speeds for CLCs that were originated in 2010 and have since converted to PLCs ("Converted PLC" red line) vs. actual speeds for PLCs originated in 2010 that were never CLCs ("Original PLC" blue line).

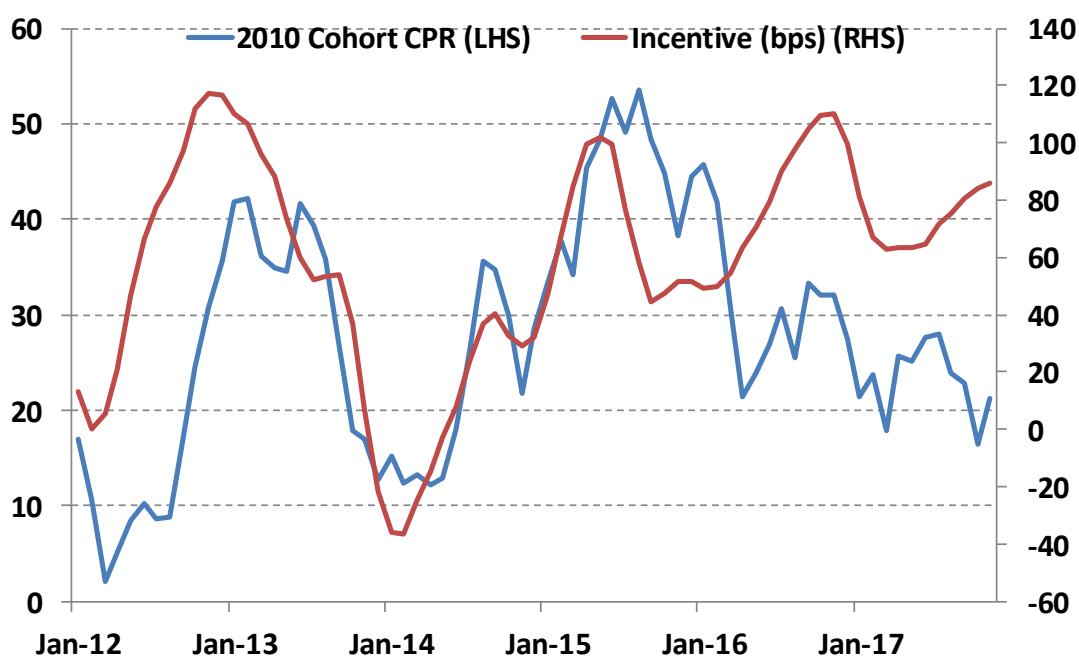
Source: Ginnie Mae, Citi.

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Burnout

If a loan has had multiple opportunities to refinance but hasn't done so there's most likely something about the loan that prevents it from being able to take advantage of such opportunities. The 2010 cohort displays this burnout effect, post-lockout, from 2016Q2 onwards. One can see in Figure 22 that even though the incentives reached 2015 levels CPRs didn't follow and instead exhibit a decreasing trend.

Figure 22. Burnout in the 2010 Cohort post-lockout



The 2010 cohort had a WAC of 4.8% at issuance. Numbers are 3-month moving averages.
Source: Ginnie Mae, Citi.

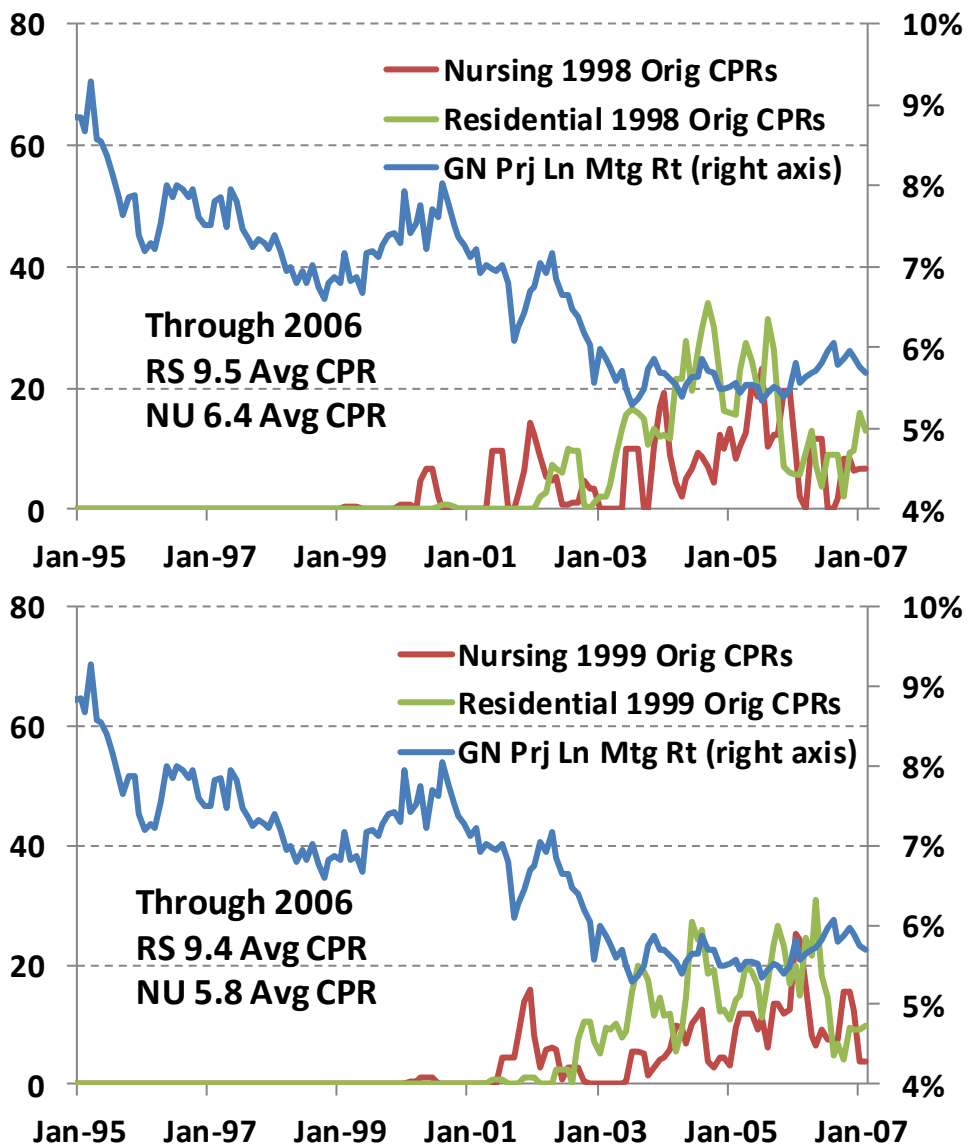
Property Type – Residential vs. Healthcare

Historical residential project loans have appeared to pay faster than nursing (healthcare excluding hospitals) (Figure 23). Residential properties are somewhat more likely to be sold than nursing properties. This is likely related to:

- a) stronger property appreciation rates for residential properties
- b) nursing properties being somewhat more specialized and as a result less liquid

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Figure 23. Residential Proj Ln Speeds Often Faster than Nursing Proj Ln Speeds – 1998 Orig. (Top), 1999 Orig. (Bottom)



Note CPRs are 3-month moving averages.

Source: Ginnie Mae, Citi.

Recent speeds, however, present a more mixed picture. Figure 24, presents recent speeds for vintage 2011-16 cohorts of Residential and Healthcare project loans. In some Residential pays faster while in others Healthcare pays faster. Hence, the model does not use the distinction between these types of project loans, for the time being.

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Figure 24. Residential vs. Healthcare Project Loan Speeds – 2011-16 Vintages

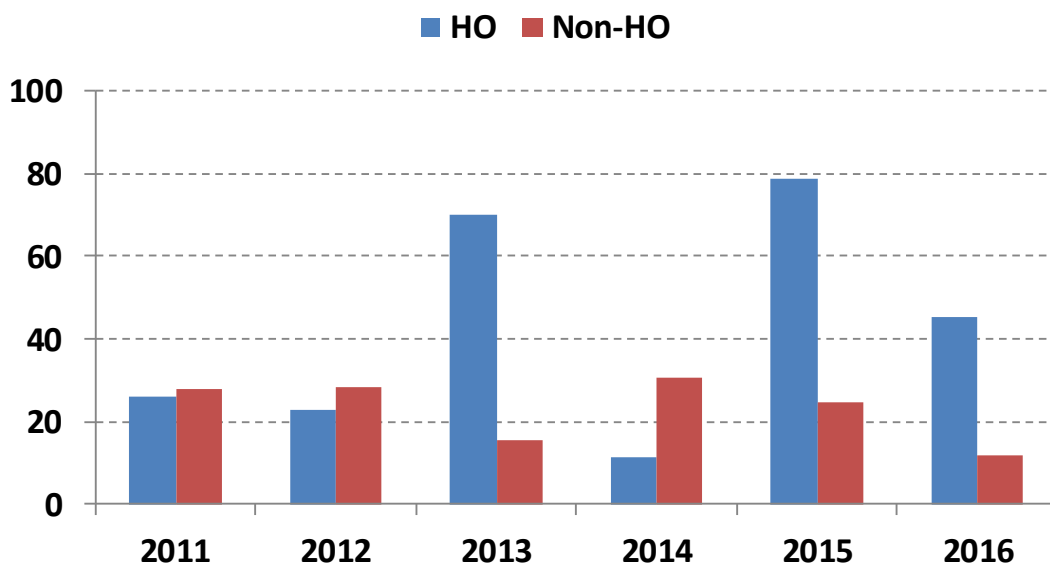
Vintage	Type	OLS (M\$)	WAC	Life CPR	12M CPR	6M CPR	3M CPR
2016	Residential	9.11	3.53	1.9	1.9	1.6	1.4
	Healthcare	9.51	3.48	0.0	0.0	0.0	0.0
2015	Residential	8.09	3.72	6.0	7.7	4.2	2.5
	Healthcare	9.04	3.74	4.1	5.7	4.7	4.4
2014	Residential	8.7	4.09	9.9	14.9	7.2	5.0
	Healthcare	8.42	4.02	10.8	17.8	11.3	12.4
2013	Residential	9.28	3.43	4.9	9.1	8.9	8.9
	Healthcare	7.99	3.62	6.6	9.4	4.5	4.1
2012	Residential	9.46	3.21	4.1	9.0	8.2	9.7
	Healthcare	7.83	3.15	4.0	8.3	6.3	7.9
2011	Residential	9.51	4.27	11.8	19.4	24.4	29.3
	Healthcare	8.48	4.29	11.3	23.9	7.8	2.4

Source: Ginnie Mae, Citi.

Hospitals

While there are hospital project loans, they are not allowed in REMICs due to the IRS requirement that the loan needs to be at least 80% real estate based and hospital loans are often backed by “operations of the facility” in addition to the underlying real estate.

Figure 25. Hospitals vs. Non-Hospital Project Loan Speeds – 2011-16 Calendar Year CPRs



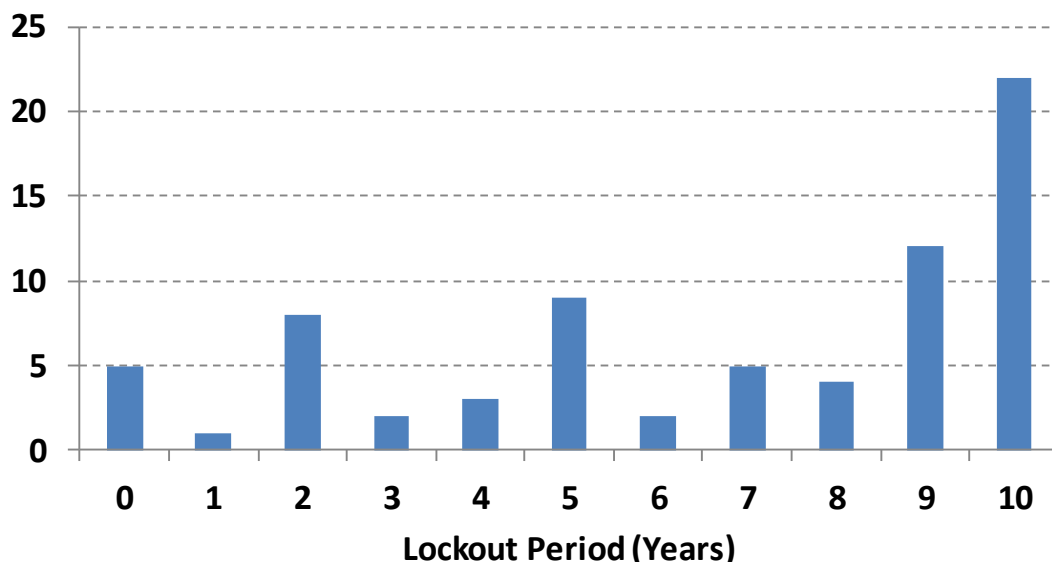
Source: Ginnie Mae, Citi.

Figure 25 demonstrates that hospitals are much more volatile. To start with there

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have historically been very few hospitals in the program, only about 170 in all. Secondly hospitals tend to have long lockout periods. Of the 73 hospital pools that are active today, 56 are currently locked. The distribution of lockouts for these 73 pools is shown in Figure 26.

Figure 26. Distribution of Lockout Periods for Currently Active Hospital Pools



Source: Ginnie Mae, Citi.

Assumability

Like FHA/VA single-family mortgages, project loans are assumable with a small assumption fee, which increases extension risk. For single-family, hassle costs in combination with strong home price appreciation can make it not worthwhile to assume an existing mortgage if one needs additional financing. For much larger project loans, the larger absolute value of the potential savings suggests assumability could be a more important factor for project loans originated during a period of low rates, such as the second half of 2012 through to early 2013.

In fact, conversations with industry participants support the view that FHA project loan financing – a long-term low-rate mortgage without recourse – is a valuable asset and that the assumability option may be exercised. However, because the steps involved are not widely known and property appreciation could render the existing FHA project loan too small, this may not come to pass.

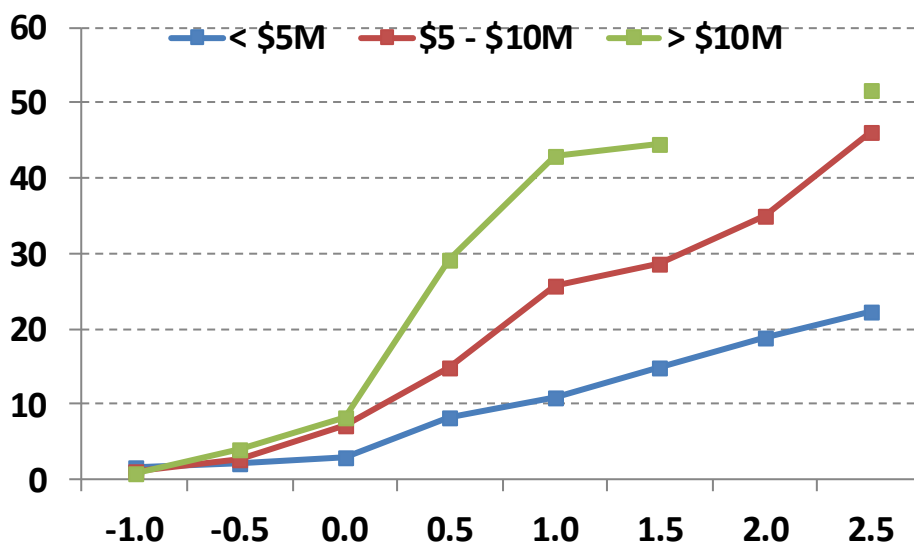
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Project Loan Prepayment Model

Penalty Adjusted Incentive vs. Loan Size

Rate incentive, prepayment penalty and loan size are major drivers of prepayment speeds. The model uses a penalty adjusted rate incentive.

Figure 27. Penalty Adjusted Rate Incentive – Loan Size Actual CPR Relationship



Source: Ginnie Mae, Citi.

High loan sizes are much more reactive than those of lower loan sizes. The model uses a modified version of the S-curves in Figure 27 with some smoothing applied to account for some of the kinks that can be seen in the actual values above.

Voluntary Prepayment Model Formulation

Prepayments are modeled as the sum of the voluntary and involuntary prepayments. The voluntary component is a product of multiple factors:

$$\text{Voluntary Prepayment} = f(\text{Core Refi}, \text{Prop Appr}, \text{Lock Exp}, \text{Pen Exp}, \text{Media Eff}, \text{Burnout}, \text{Prop Type})$$

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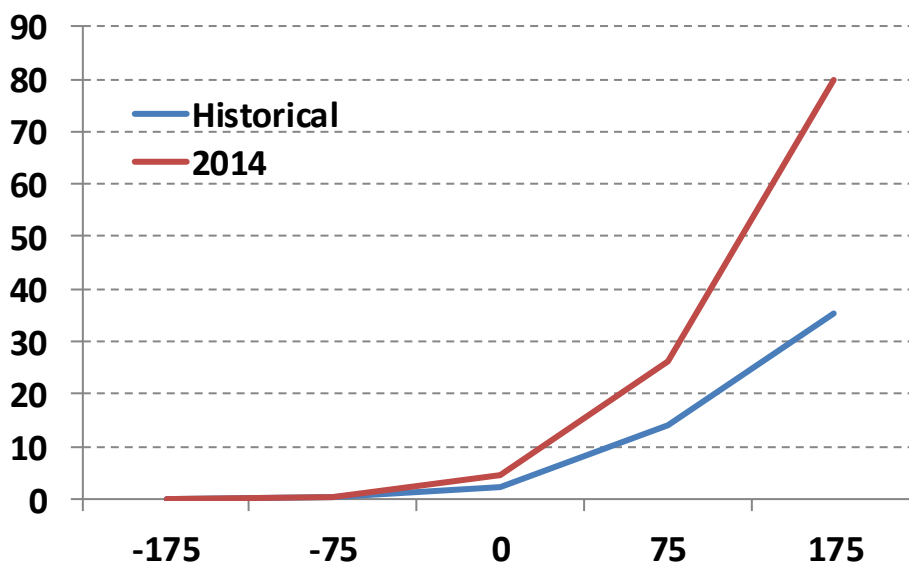
Estimation of Incentive

At a theoretical level, the dollar incentive is:

$$\text{Dollar Incentive} = (\text{WAC} - \text{Penalty Adjusted Project Loan Rate}) * \text{Loan Size}$$

While the theoretical estimation may be correct, in practice the decision to prepay appears to be significantly more complex than the simple expression would imply. Figure 28 shows that ITM High Loan Size loans, with prepayment penalties of 7-10%, have prepayment speeds in 2014 that are twice the historical average. This is likely due to a combination of a period of really low rates, property managers administering with greater *nous* and aggressive brokers turning over every rock and looking under every seat cushion for more business.

Figure 28. High Loan Size and High Penalty Loans on a Recent Prepayment Sugar Rush



Source: Ginnie Mae, Citi.

Project Loan Defaults

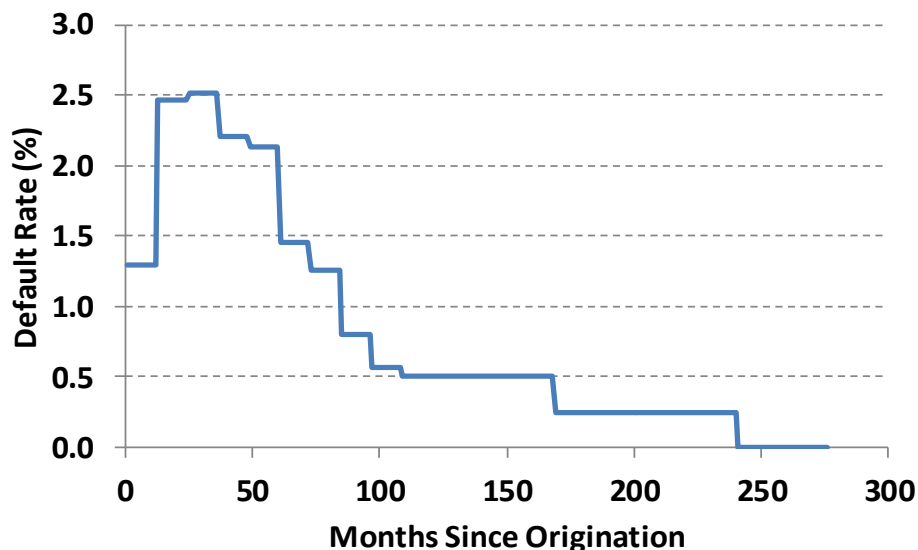
FHA project loans naturally can default like other mortgages, although FHA insures the loan and Ginnie Mae further guarantees loans securitized into pools. Together these guarantees protect the investor's principal and interest payments.

A standard default curve called the Project Loan Default or "PLD Curve" is used by

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the industry to gauge default behavior with loan age (Figure 29).¹⁸

Figure 29. Standard Project Loan Default (PLD) Curve



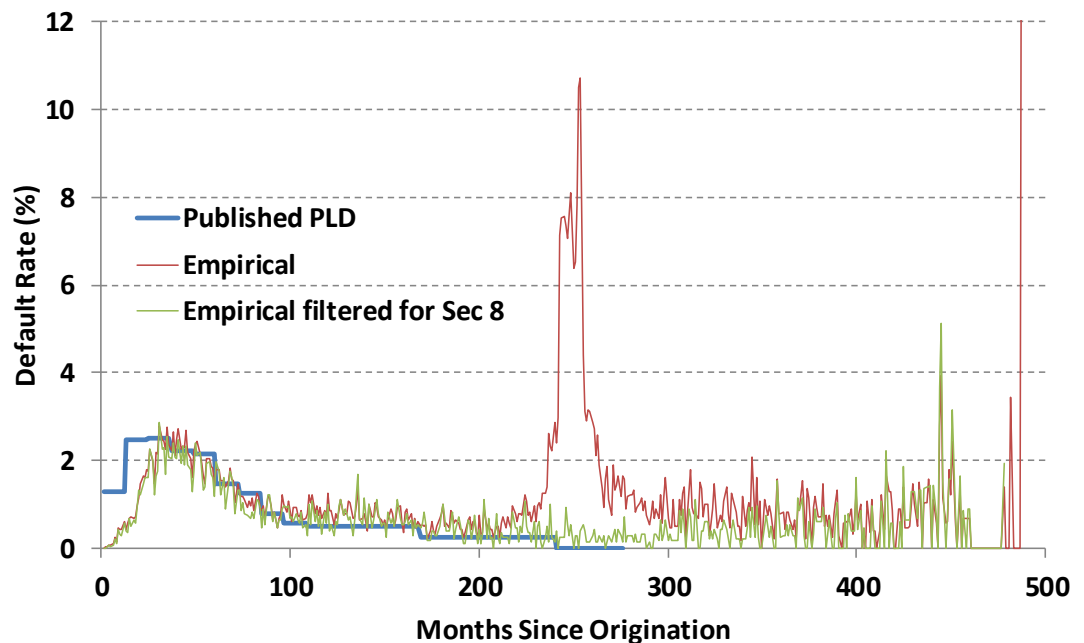
Source: DLJ, Citi.

And although it appears to be generally in agreement with actual default data shown in Figure 30, the PLD Curve (in blue) appears to be somewhat high for new project loans (during the first 2 years) and somewhat low for very seasoned (20+ years old) loans.

We believe that the low level of actual defaults (red and green curves) for the first two years are related to the high levels of refinancing seen in recent years, especially in 2012-2013. We have observed that, even for construction loans, which are typically risky, the default rate is low (consistent with our empirical PLD) in the years. However, even when only using relatively high default 2002 and earlier data, there remains a sizeable gap between the PLD curve and actual data.

¹⁸ The origin of the PLD curve is believed to be Donaldson, Lufkin & Jenrette (DLJ) Securities Corp – a supplement document for the DLJ project loan deal Fannie Mae REMIC 1998-M7 states that the PLD curve was “developed by the Dealer”. Another source incorrectly suggests that the PLD curve was developed more recently than 1998. Furthermore, in the same DLJ deal document, a voluntary prepayment “PLS” was given, but apparently the market instead adopted the current CPJ convention. CPJ is defined as the standard PLD curve plus CPR (so “15 CPJ” = 15 CPR plus defaults at PLD[Age]).

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Figure 30. FHA Project Loan Default Curve and Actual Default Data


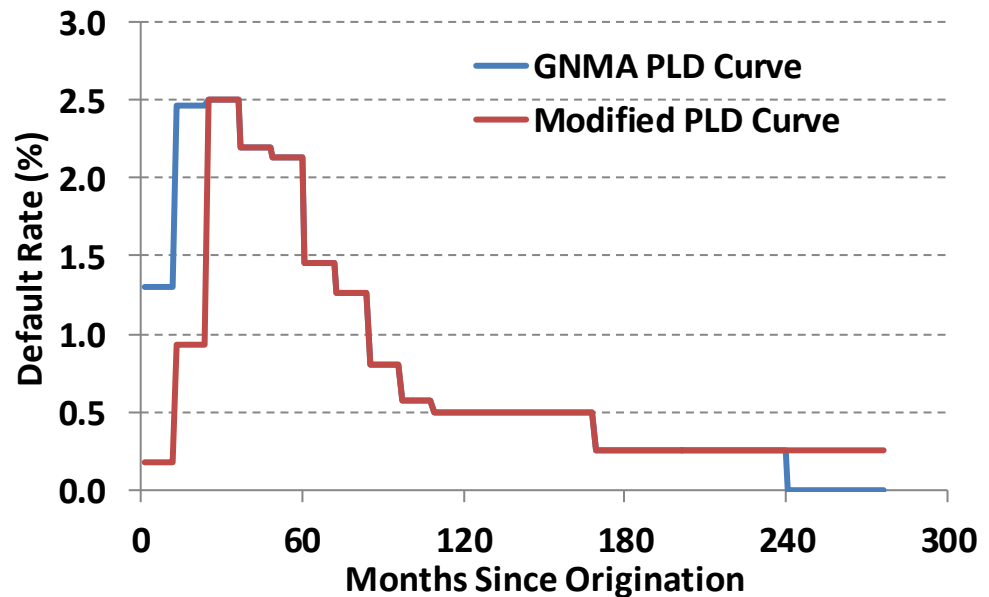
Source: HUD, Citi.

Another idiosyncrasy in the empirically derived default rate data occurs in the form of a very large peak at the 20 year mark (Figure 30 red data). We found this to be due to FHA “New Construction and Substantial Rehabilitation” (Housing Act Sections 221(d)(3) and 221(d)(4)) loan pools for apartments and coops originating in 1975-1985. The default in these pools occurs approximately concomitantly with Section 8 (rental subsidizing) program policy changes. Filtering the available data for such loans results in the expected PLD curve behavior (Figure 30 green curve).

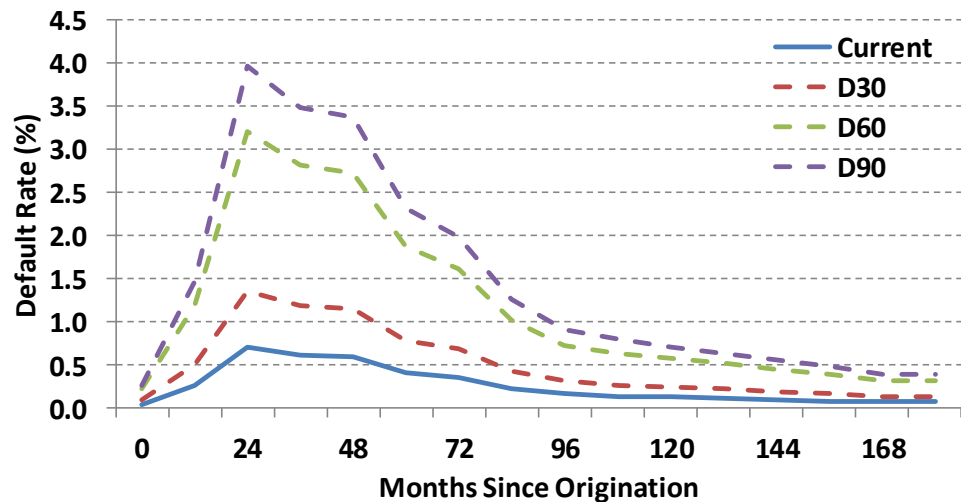
Given the above considerations, it seems difficult to justify using the standard PLD curve. The “modified” PLD curve, based on empirical observation, starts with very low default in the first year, rises more quickly initially, and additionally, remains flat at 0.25% after year 20 (instead of going to zero) (Figure 31a). The model uses a smooth curve rather than a step function (Figure 31b) as discrete changes are less likely than smooth ones.

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Figure 31. Standard and Modified PLD Curve Comparison



b)



Source: DLJ, Citi.

Delinquencies

The modified PLD curve, like the original, is applicable to loans that are not delinquent at the time of projection. For delinquent loans at the time of projection, an adjustment is required.

Figure 32a shows that default rates have been under 1 CDR for a number of years, including loans that have been delinquent during that period of time. A multiplier that boosts the PLD curve – a greater boost for a longer delinquency – seems

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reasonable. The scaled PLD curves for delinquent loans are also shown in Figure 31b.

LTV

While defaults are often considered a function of LTV, it should be noted that defaults actually declined slightly as property values fell during the post housing bubble, property value collapse (Figure 32a shaded area). This trend was not due to the FHA providing refinances and loan modifications at a fast pace, which has only become the case over the last three years. Rather, during the early to mid-2000s, the FHA had experienced credit losses, in part, due to a number of so-called “contrived defaults”, which inflated the number of observed defaults during that time period (cf. the larger default rates from 2000-2006, Figure 32a).

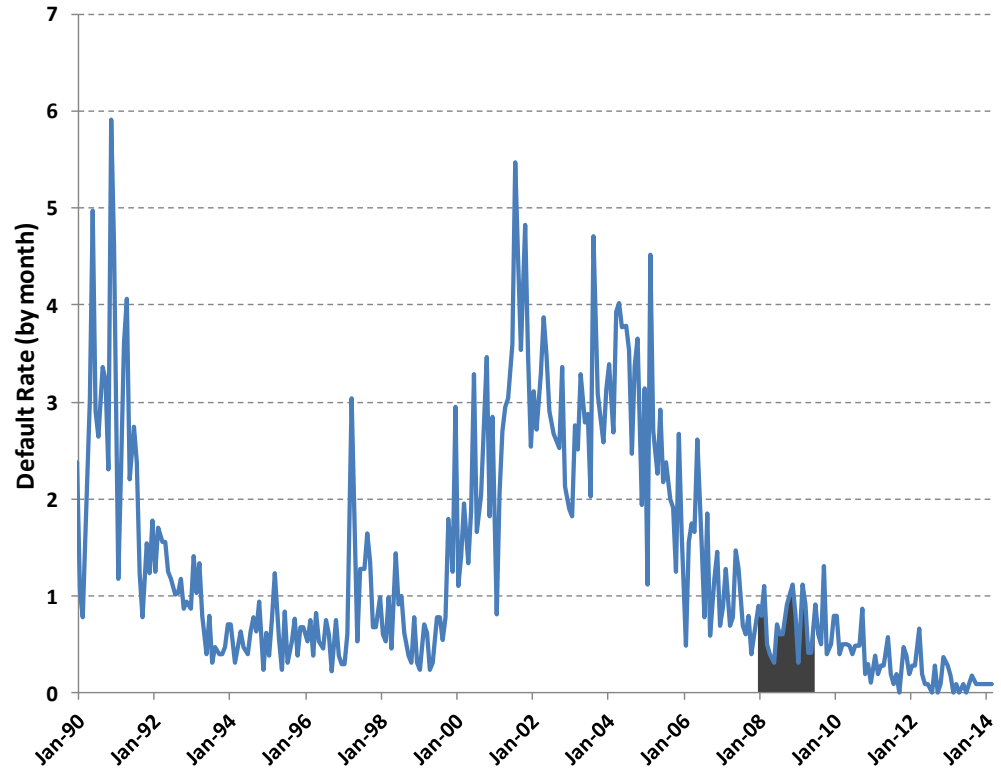
In such contrived defaults, distressed borrowers and servicers would conspire to obtain HUD approval to relax the call protection provision and/or lower the loan interest rate by arguing that the property had underwent substantial deterioration in value. Subsequently, the borrowers would negotiate with the HUD to void (or “override”) provisions for call protection, the Ginnie Mae pool would be paid off without penalty, and the property would be refinanced into a new project loan. This loophole in protocol has since been closed, and HUD now moderates all loan modification/override-type negotiations.

During the post-housing bubble, the FHA portfolio performed well, and as the multifamily market improved, defaults have lowered each year since. Currently, the FHA portfolio is extremely large leading to a surprisingly low default rate and borrowers with low interest loans.

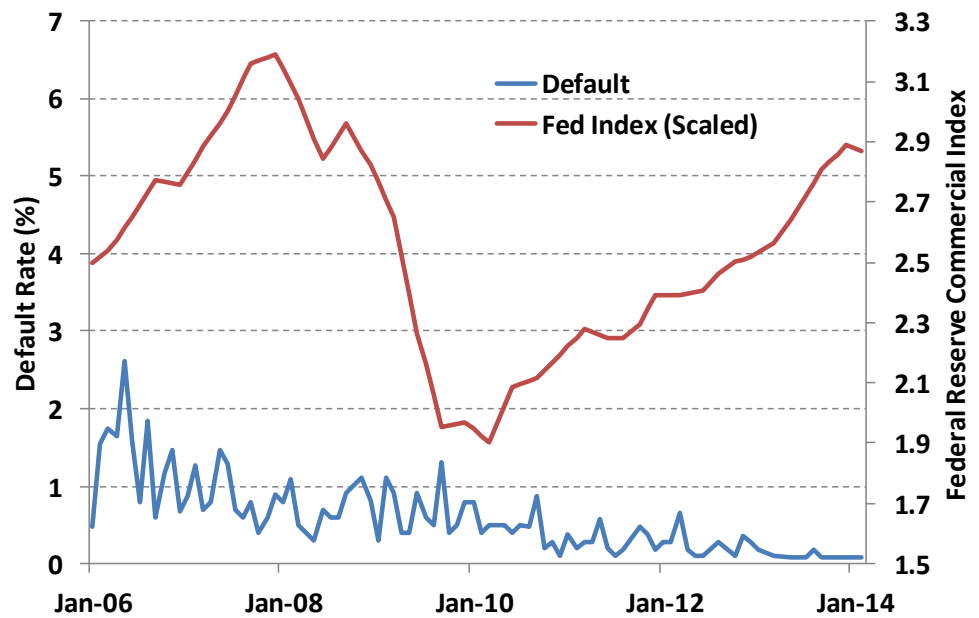
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Figure 32. Project Loan Default Rate over Time

a)



b)



Source: HUD, Federal Reserve, Citi.

As seen in Figure 32b, there has been no clear discernable trend in the rate of

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default with property appreciation as gauged by the Federal Reserve Board (FRB) Commercial Property Index. Therefore, we turn to sources, other than LTV, to model default dependence on appreciation. It should also be noted that LTV is not provided by either HUD or GNMA.

Involuntary Prepayment Model Formulation

The involuntary prepayment model is the modified PLD curve, boosted by the state of delinquency at the time of projection.

$$\text{Involuntary Prepayment} = f_{MPLD}(\text{Loan Age}) * f_{Delinquency}(\text{Delinquency State})$$

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Applications – OAS and Duration

Given that Project Loans clearly react to refinancing incentive, a flat CPR/CPJ assumption probably makes less sense when estimating valuations in many interest rate scenarios as is done for OAS analysis.

Prepay Projections Across Pools and Scenarios

In Figure 33 the model has been run for each of the pools backing deal GNMA12.139. The pools are ordered by WAC, which more or less corresponds to an ordering by CPR.

The model brings out other subtle differences. Pools AB2574 and 777739 (red circles) have the same WAC but very different loan sizes. This explains the difference in projected speeds should the rates rally by 1%. Taking this one step further pools AA7807 and 793926 (purple circles) have similar WACs but the latter pool is larger by half. So why does 793926 have the smaller 1-Yr CPR? Because it still has a few more months of lockout, after which it will face a 9% penalty. AA7807 has been unlocked for a few years and is facing a marginally lower penalty for the year. Thus, a constant CPR/CPJ projection across all pools, while simpler, is likely to be less accurate.

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Figure 33. GNMA 12.139 Collateral – Projected 1-Yr. CPR and Scenario LT CPRs

Pool	Coupon	WAC	OLS	1-Yr CPR	LT CPR Scenarios		
					-100	0	100
791423	4.10	4.35	13.1	21.3	88.1	21.1	7.4
AA1502	3.85	4.35	5.6	17.3	64.7	15.6	6.5
AC0545	3.50	3.75	11.9	7.7	41.3	9.5	3.8
428083	3.47	3.72	9.5	5.6	35.1	8.5	3.6
AA7807	3.40	3.65	9.7	7.0	32.2	8.4	3.8
793926	3.35	3.60	14.3	5.1	32.9	8.0	3.7
AB2574	3.20	3.50	16.3	5.9	22.0	7.2	3.9
777739	3.25	3.50	1.6	3.5	9.6	4.3	2.0
768187	3.20	3.45	7.1	5.6	16.6	6.6	3.6
791447	3.05	3.40	8.9	7.1	20.8	7.1	4.4
778468	2.60	3.20	2.2	3.5	10.9	4.1	2.6
AB2577	2.76	3.14	2.0	2.8	9.6	3.3	2.4
AB2576	2.76	3.14	1.6	2.3	7.6	2.6	1.9
AB2579	2.62	3.05	5.6	3.1	11.7	3.8	3.3
791914	2.78	3.03	4.3	3.0	11.5	3.6	3.2
AB0880	2.74	2.99	63.2	0.6	10.2	3.5	3.4
AA9254	2.60	2.98	13.6	3.1	12.1	3.9	3.8
AA9768	2.45	2.95	7.8	2.8	10.9	3.6	3.6
AC1401	2.45	2.70	3.7	2.4	8.5	3.1	3.1
AA2039	2.40	2.65	57.0	2.6	8.8	3.8	3.8
AA7814	2.15	2.65	11.5	2.5	8.6	3.7	3.7
AB2566	2.40	2.65	4.7	2.4	8.1	3.2	3.2
AB0876	2.39	2.64	30.7	0.6	7.2	3.4	3.4
798588	2.33	2.58	7.1	2.4	7.6	3.5	3.5
AA7780	2.17	2.55	3.7	2.4	6.9	3.0	3.0
793418	2.17	2.50	25.3	2.4	7.1	3.8	3.8
793420	1.99	2.49	2.1	2.2	5.2	2.4	2.4
AA8463	2.23	2.48	3.7	2.5	7.0	3.3	3.3
745572	2.15	2.45	6.4	2.5	6.5	3.5	3.5
745571	2.15	2.45	5.3	2.5	6.4	3.3	3.3
750451	2.20	2.45	4.8	2.5	6.8	3.5	3.5
745573	2.15	2.45	4.6	2.5	6.3	3.2	3.2
AB7104	2.10	2.44	5.6	2.4	6.3	3.3	3.3
745567	2.10	2.40	9.9	2.5	6.3	3.7	3.7
750450	2.15	2.40	2.7	2.4	5.8	2.9	2.9
798596	2.13	2.38	3.6	2.4	5.8	3.1	3.1
750453	2.10	2.35	2.7	2.6	5.6	3.0	3.0
AA3415	1.97	2.22	10.6	0.6	4.3	3.2	3.2

Source: Ginnie Mae, Yield Book, Citi.

Given that the Project Loan Prepay Model handles the nuances of project loans, a

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benefit to using it is that OAS and duration measures can be computed in the same way as they are for single-family residential MBS.

Figure 34 demonstrates this capability by comparing two of the pools shown in Figure 33 above. The two pools are identical in every way except that one has a slight higher loan size than the other. This higher loan size translates into a slightly higher price, which is likely due to a greater likelihood of collecting a penalty payment.

Figure 34. Pool Comparison – Projected Prices and Duration Measures at -25 OAS

Pool	AB2577	AB2576
Coupon	2.76	2.76
WAC	3.14	3.14
OLS	2.0	1.6
WAM	319	319
Age (Yrs.)	2.7	2.7
Lockout	Dec-12	Dec-12
Penalty	Dec-22	Dec-22
Price	101.10	100.94
EffectiveDuration	7.8	8.2
EffectiveConvexity	-0.5	-0.5
WAL	11.3	11.9
OneYearCPR	2.8	2.3
LongTermCPR	3.3	2.6

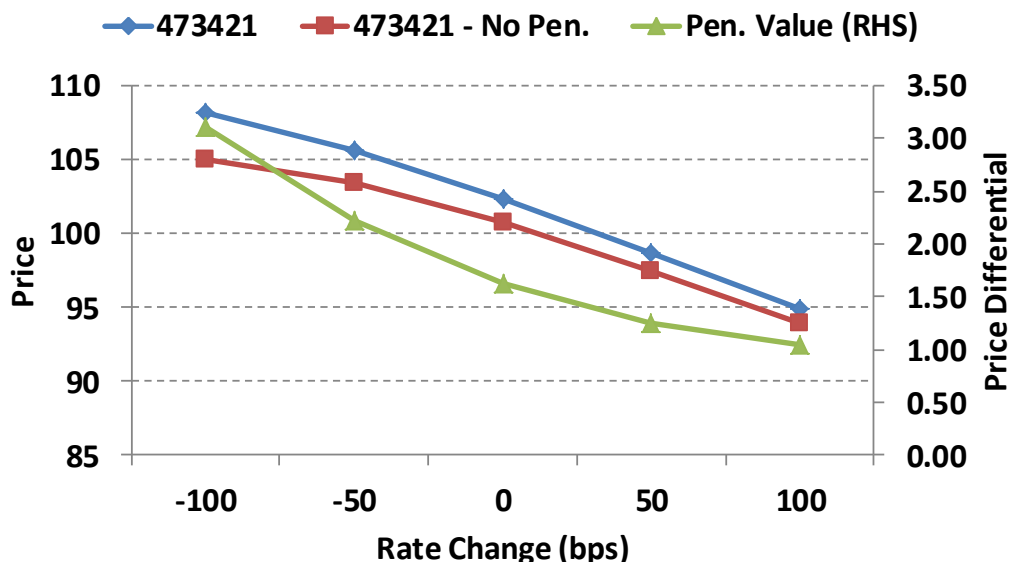
Source: Ginnie Mae, Yield Book, Citi.

One can also use the model to consider the impact of the penalty, as demonstrated in Figure 35. The price of Pool 473421 is charted according to different interest rate scenarios, both with (blue line) and without the penalty (red line). The difference in prices is the value of receiving the penalty cash flow. As one would expect, the greater the chance of prepayment the more valuable the prepayment penalty cash flow.

Convexity – We can also observe the impact of the penalty cash flow on effective convexity. Normal pool 473421 has an effective convexity of -0.8. But if the upside of receiving the penalty cash flow is removed, then the effective convexity becomes a more negative -1.7 (since the penalty cash flow is more likely to be received in a rally scenario in which speeds rise).

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Figure 35. GNMA 473421 with and without Penalty in Immediate Rate Shift Scenarios (-25 OAS)



Source: Ginnie Mae, Yield Book, Citi.

From the preceding, it is clear that Ginnie Mae Project Loan collateral, while certainly “lumpier” than residential single-family collateral, shares call risk and extension risk characteristics of single-family collateral.

For the Unconvinced – Model Override

Given the relatively small number of loans in deals and diversity of opinion on prepayments, Yield Book® not only allows the user to not just override the Ginnie Mae Project Loan model, but to override it at the loan-level.

While in theory better than using some simple rules of thumb, prepayment models are “black boxes” – meaning that, at times, the projections can seem counter-intuitive and require some time to understand. We hope that the “model override” feature on Yield Book® will help users to grow comfortable with model projections by allowing them to “turn the model on” gradually, even one loan at a time if desired.

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Figure 36. Yield Book GN Project Loan Model Override Feature (Right-Middle Part of Screen)

First 2 pools (ordered by balance) run at 15 CPJ, others run at 100% of prepayment model.

Source: Ginnie Mae, Yield Book, Citi.

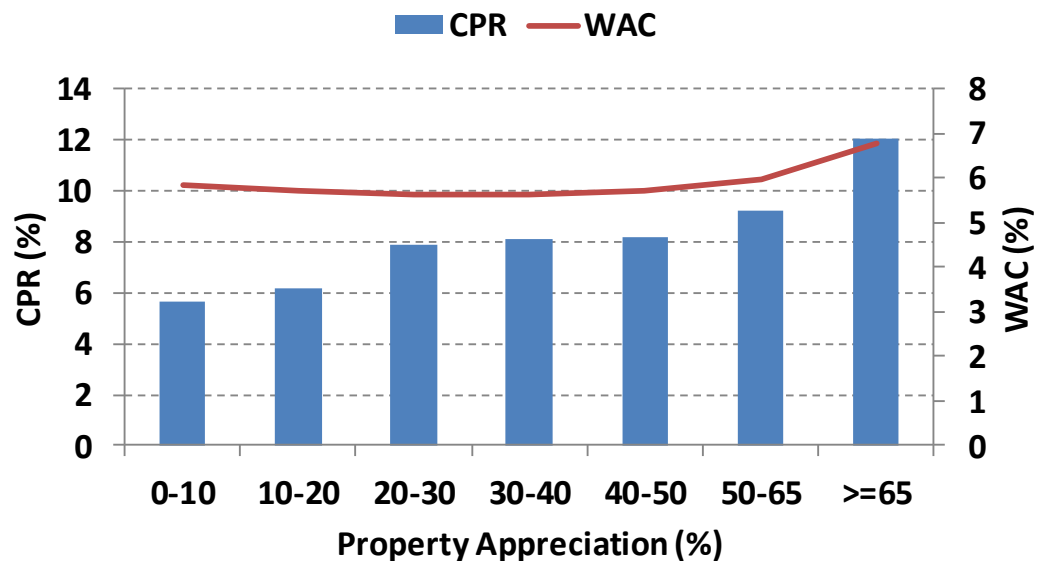
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Future Considerations

Property Appreciation

Initial analysis, shown in Figure 37, shows a weak relationship between property appreciation and prepayment speeds. While speeds do increase with appreciation, they only do so from 6 CPR for 0-10% appreciation to 9 CPR for 50-65% appreciation.

Figure 37. Property Appreciation vs. Prepayment Speeds

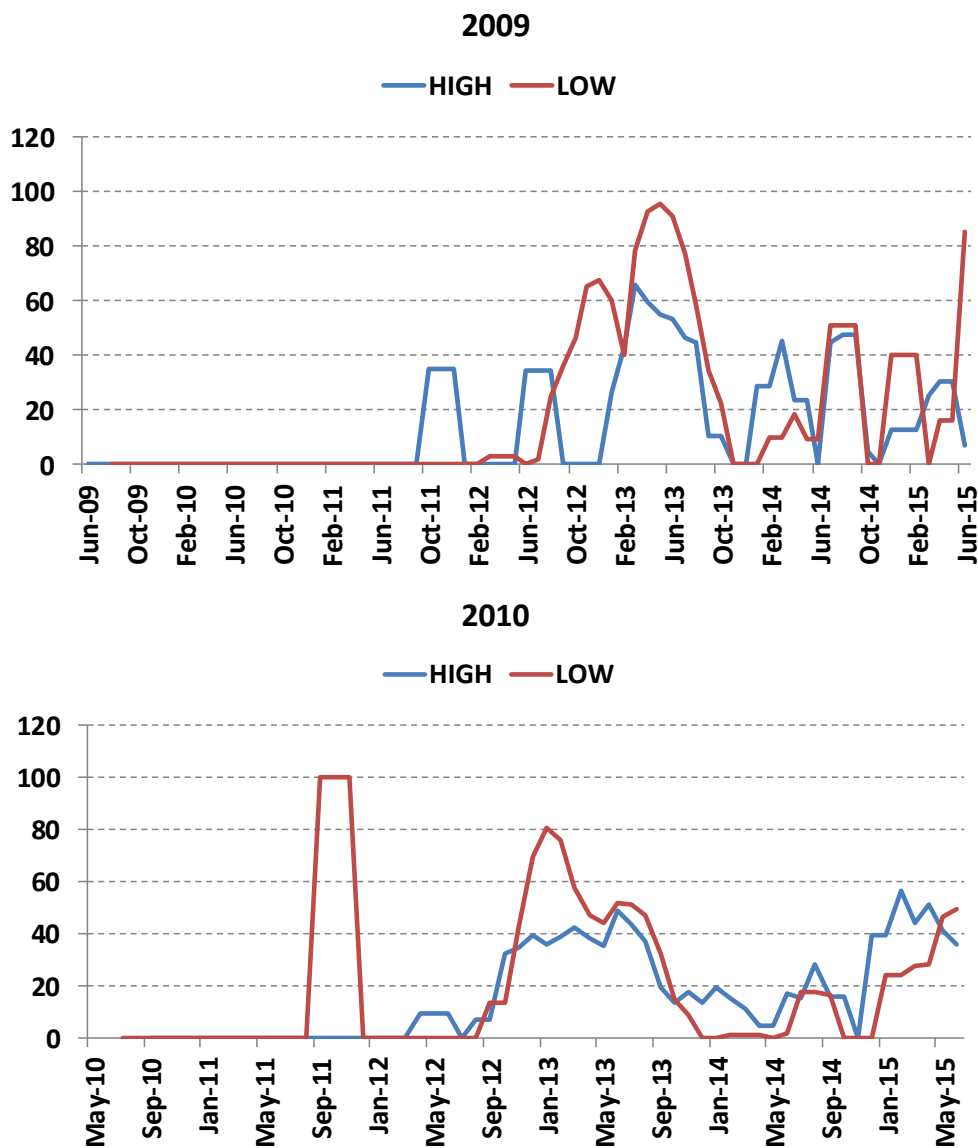


Source: Citi.

For further analysis, we took a particular vintage and divided its pools into those that experienced high and low appreciation. Figure 38 further supports the belief that the relationship between property appreciation and prepayment speeds is weak, at best.

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Figure 38. Prepayment Speeds by Vintage and Property Appreciation Level (High and Low)



The 2009 cohort has WACs of 5.72% (High) and 5.63% (Low)

The 2010 cohort has WACs of 4.94% (High) and 4.39% (Low)

Source: Ginnie Mae, Citi.

Single-Family Experience – While it was difficult to observe the impact of property appreciation on project loan speeds, the much larger single-family prepayment data set provides an alternative perspective. Controlling for loan size, single-family property appreciation bucketing suggests a substantial property appreciation effect of as much as 20% faster speeds per 10% accumulated property appreciation. Naturally, multi-family property managers do not necessarily behave the same way as single-family mortgage borrowers, but this result does suggest that this is an

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area of further work.

Programs

The FHA has a number of different programs under which insurance for a project loan can be obtained. We considered the 4 most common to look for differences in prepayment speeds:

207/223(f) – insures mortgage loans to facilitate the purchase or refinancing of existing multifamily rental housing

232/223(f) – allows for the purchase or refinancing of nursing homes, intermediate care facilities, board and care homes, and assisted-living facilities with or without repairs of existing projects not requiring substantial rehabilitation

221(d)(4) – insures mortgage loans to facilitate the new construction or substantial rehabilitation of multifamily rental or cooperative housing for moderate-income families, elderly, and the handicapped

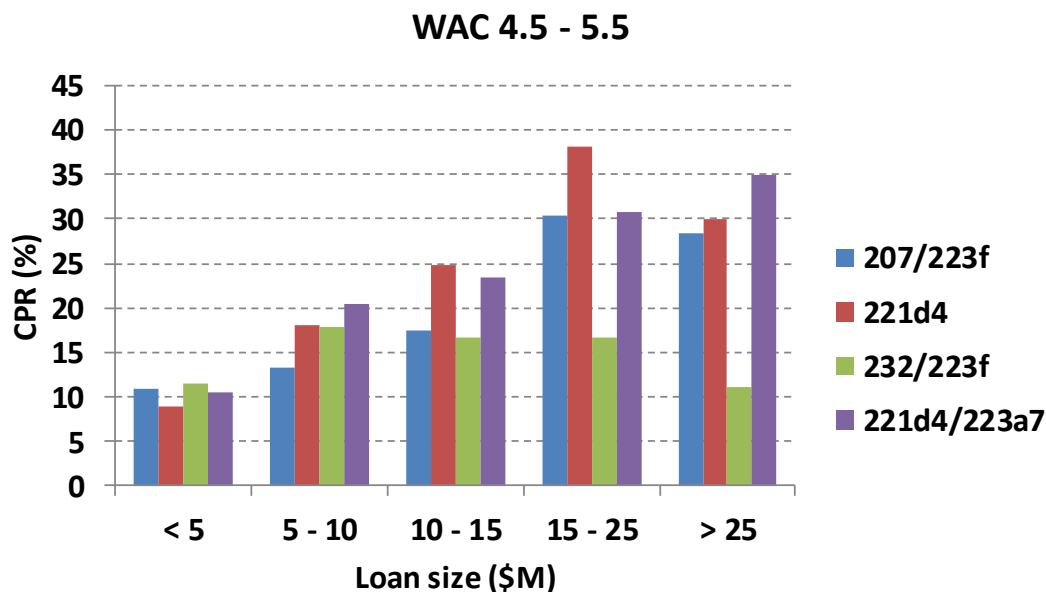
221(d)(4)/223(a)(7) – permits refinancing of mortgages on multifamily projects already insured by FHA

Figure 39 shows some weak relationships based on the program. For example, we would expect nursing homes – 232/223(f) – in general to prepay slower than other programs and the green columns do appear to be smaller than the others. It is possible that properties that have refinanced once – 221(d)(4)/223(a)(7) – are run by savvier property managers who are more aware of opportunities to refinance and hence, we may expect them to refinance more often. It isn't clear from the chart whether they really do prepay faster than those properties that haven't refinanced at least once – 221(d)(4). The higher speeds of the 221(d)(4) pools could be due to the recent spate of IRR/LM refinances.

Segmentation by program warrants future exploration.

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Figure 39. Property Appreciation vs. Prepayment Speeds



Source: Ginnie Mae, Citi.

Appendix

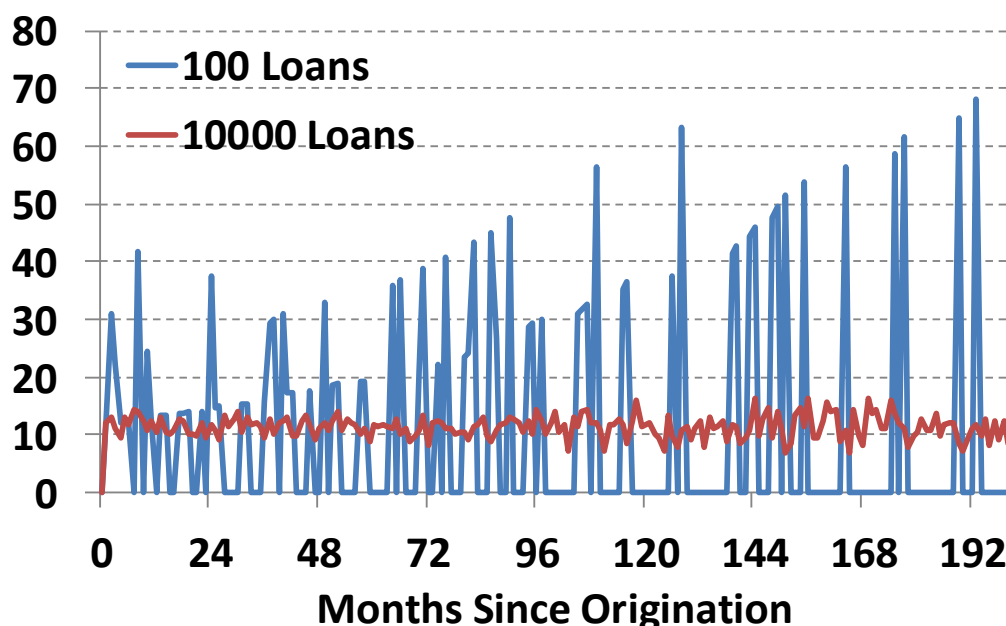
Appendix I – Deal “Sample” Size and Volatility

Relative to single-family residential housing, project loan deals typically involve many fewer loans. As a result, there is naturally more prepayment “noise” associated with project loan deals. To illustrate the difficulties inherent in project loan prepayment modeling, we consider a simple toy model of prepayment for two separate pools of identical loans, one consistent with the size of a project loan CMO deal ($N_{pool} \approx 100$ loans), and the other consistent with the size of a residential single family housing CMO deal ($N_{pool} \approx 10,000$ loans). We assume that 1% of all outstanding loans are fully prepaid on average each month. Drawing loans (1% of N_m loans on average; N_m = number of surviving loans in month m) for termination each month, from a binomial distribution, we obtain the CPR of each deal over time.

The simulation results in Figure 40 show what may be intuitively obvious – all other things being equal – the smaller the deal size, the larger the dispersion in CPRs. This, in tandem with the inherent difficulties in modeling prepayments, makes the modeling task in the case of project loans a particularly thorny one.

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Figure 40. Simulations Illustrating Sample Noise for a 100 Loan Deal vs a 10,000 Loan Deal



Roughly speaking, the decline in standard deviation is inversely proportional to the number of loans. SMM is assumed to be a constant 1%.

Source: Citi.

Appendix II – Major Project Loan Issuers

The issuance market tends to be pretty concentrated at the top. In any given year the top 10 years will typically account for about two-thirds of all pool issuance for the year.

Figure 41. Major Project Loan Issuers in 2016

Issuer	Issuance (\$M)	%
BERKADIA COMMERCIAL MORTGAGE, LLC	1,767	11%
P/R MORTGAGE AND INVESTMENT CORP	1,384	9%
RED MORTGAGE CAPITAL, LLC	1,342	8%
GREYSTONE FUNDING CORPORATION	1,260	8%
WALKER & DUNLOP LLC	1,238	8%
LANCASTER POLLARD MORTGAGE COMPANY	937	6%
KEYBANK NA	860	5%
WELLS FARGO MULTIFAMILY CAPITAL	662	4%
LOVE FUNDING CORPORATION	583	4%
HOUSING & HEALTHCARE FINANCE, LLC	521	3%
All Others	5,683	35%
Total	16,237	100%

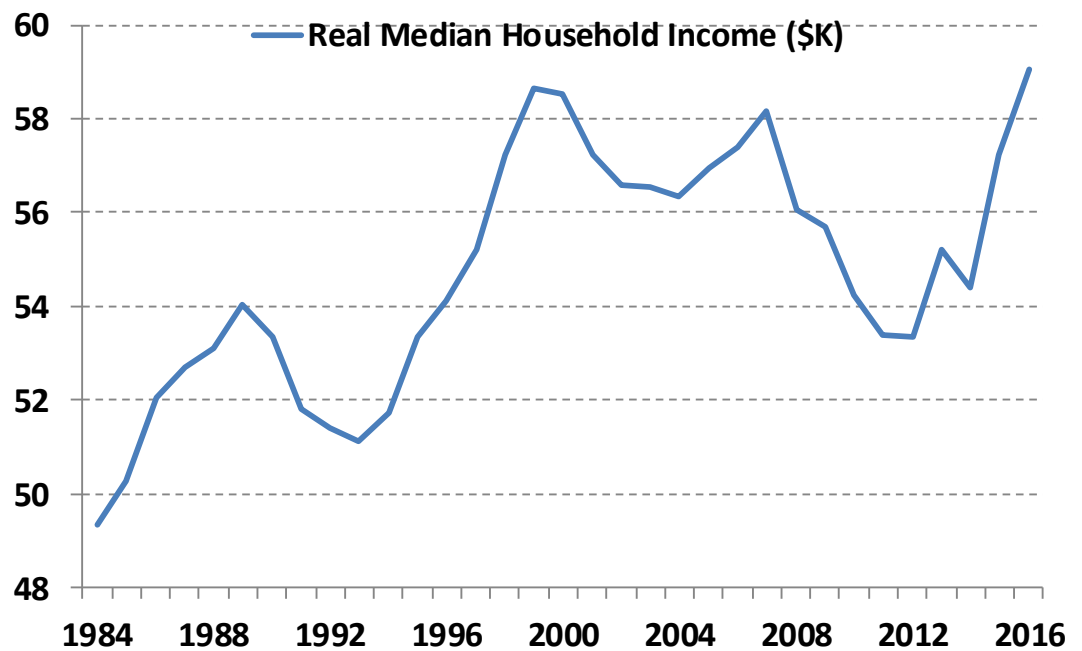
Source: Ginnie Mae, Citi.

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Appendix III – Multifamily Growth Drivers

Weak income growth and additional financial stress (for example due to student loan debt) impacting especially the middle class (as well as lower income households) made it more difficult to attain homeownership, post-recession, resulting in more renters. Median income started to rebound in 2012 and has is now at a record high.

Figure 42. Inflation-Adjusted Median Household Income Has Only Recently Begun to Rebound

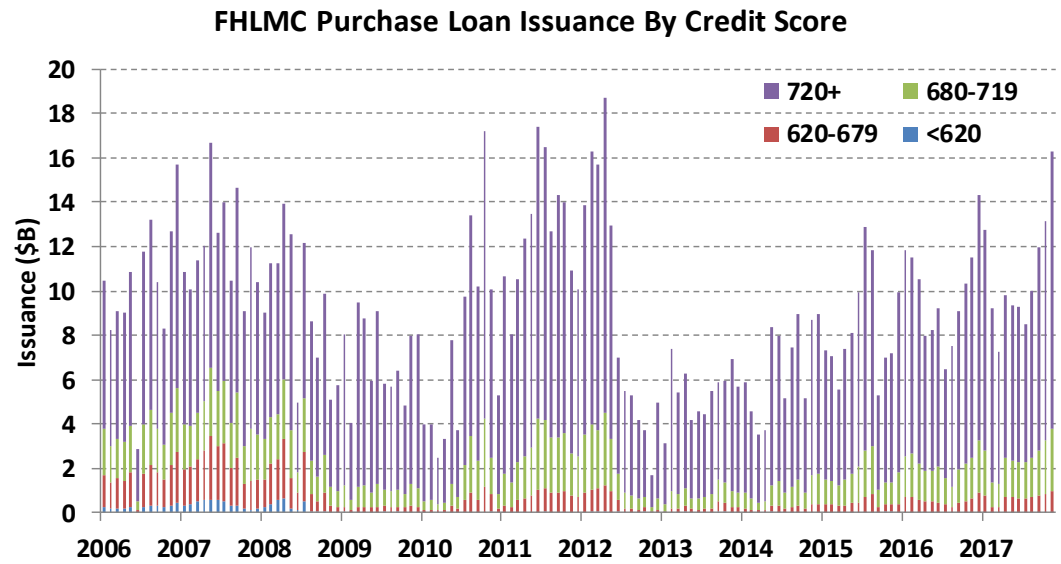


Source: Census Bureau, Citi.

Tight Mortgage Underwriting – Naturally tighter underwriting has made renting perhaps the only option for many households that were previously able to buy.

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Figure 43. Freddie Mac Purchase Loan Issuance FICO Score Distribution



Source: Census Bureau, Citi.

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